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Scotland versus the Rest:
A Fiscal Investigation into the Comparative Exposure of Artificial
Intelligence in Scotland and the Rest of the UK

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Abstract

Artificial intelligence is expected to reshape labour demand unevenly across occupations, and therefore across regions whose occupational composition differs. Whether Scotland is differentially exposed relative to the rest of the UK (rUK) is a fiscal question, yet exposure measures report point estimates without characterising their dependence on the scoring model. This dissertation makes three contributions. First, I prove that the regional differential in an employment-weighted exposure index removes the common component of any model-specific bias, that an affine bias survives only as a rescaling which a standardised or rank-based differential removes, and that the remainder is Cauchy–Schwarz bounded. Second, I answer it, and test the answer by re-scoring all 17,945 O*NET tasks on two further models. Scotland sits about two percentile points below the rUK in the UK occupational exposure distribution, or 0.86 index points on a 0–100 scale (95% interval $[-1.07, -0.69]$), with London accounting for two-fifths of that gap and Scottish exposure tilting towards substitution. The sign and percentile magnitude hold across models where the index-point figure does not, the alternatives compressing the scale rather than reordering occupations. Third, the gap implies a ten-year TFP differential under three basis points and a net-position sensitivity of about half a million pounds a year. The lesson is that scale-free differentials can be trusted where levels cannot.

Keywords: artificial intelligence, labour-market exposure, occupational composition, Scotland, shift-share, measurement error.

JEL classification: J21, J24, O33, R23.

1 Background and Related Literature

The fiscal weight of Scotland’s labour market rests on the settlement that followed the Scotland Acts of 2012 and 2016. Devolved and partially devolved taxes now fund close to half of the Scottish Government’s resource spending, with non-savings, non-dividend income tax the single largest component (Scottish Fiscal Commission 2026). The block grant adjustment (BGA) is what makes the composition of the tax base consequential. When income tax was devolved the block grant was cut by the revenue foregone, and that deduction indexed thereafter to the growth of the equivalent rUK tax base (Scottish Government and HM Treasury 2023). The Scottish budget therefore gains only insofar as Scottish income tax grows faster than its rUK comparator, so that the level of Scottish revenue is immaterial and only the differential matters. The Scottish Fiscal Commission (SFC) records this as the income tax net position, the gap between devolved revenue and the BGA, forecast at £969 million for 2026–27 (Scottish Fiscal Commission 2026). A relative deterioration in Scottish earnings, however modest in absolute terms, feeds mechanically into a smaller net position, and because outturn is reconciled against forecast three years on, an error made today constrains spending well into the future (Scottish Government and HM Treasury 2023).

It is against this backdrop that the fiscal repercussions of AI-driven labour-market disruption matter. A technology that raises productivity and reshapes labour demand unevenly across occupations will reshape earnings unevenly across the economy, and any resulting divergence between the Scottish and rUK tax bases surfaces in the net position the SFC must forecast (Scottish Parliament 2016). There is good reason to expect such a divergence, rooted in Scotland’s distinctive occupational composition, and its direction is an empirical question: the composition may harbour Scotland from displacement, or it may deny Scotland the gains that accrue to complemented work. This is heightened by a longstanding productivity gap. With growth weak across Scotland, the UK, and much of Europe since the 2008 crisis, and forecast downgrades to productivity already worsening the fiscal outlook (Scottish Fiscal Commission 2026), whether AI narrows or widens that gap turns on how its gains are distributed across regions.

The task mechanism: displacement, complementarity, and the cognitive frontier

The labour-market consequences of AI run through the task. Whether AI raises or lowers the demand for a worker depends on which of the activities that make up the job the technology can complete, complement, or cannot do. The organising framework is the task-based model of Acemoglu and Restrepo (2019), in which production allocates a continuum of tasks between labour and capital, and a technology acts through two opposing channels. A displacement effect operates as capital absorbs tasks workers performed, and a reinstatement effect operates as new tasks restore a comparative advantage for labour. The central lesson is that productivity gains and labour demand can move apart, and Acemoglu and Restrepo (2020) show that “so-so” automation, which displaces workers while delivering only modest productivity gains, can depress employment and the labour share even as output rises. Exposure to a technology is thus not the same as harm from it: a task AI can perform is displaced, whereas a task AI makes a worker

faster at is complemented, and the two carry opposite signs for labour demand.

Which tasks a technology reaches has changed. Earlier computerisation acted on routine, codifiable tasks, the empirical content of the routine-biased technical change literature (Autor et al. 2003; Autor and Handel 2013), hollowing out middle-wage clerical and production work while sparing both manual and analytical non-routine activity. Generative AI reaches the non-routine cognitive tasks the previous wave could not, so exposure no longer maps onto the routine/non-routine divide that organised two decades of work. The theoretical fault line is consequently the substitution–complementarity margin, and the literature divides on which dominates. Autor (2024) reads AI optimistically, as a tool that could reinstate middle-skill judgement and widen access to expert tasks, whereas Acemoglu (2025) is more cautious, projecting an aggregate total-factor-productivity gain of under one per cent over a decade once only currently automatable tasks are counted. The disagreement concerns magnitude and net sign, not the mechanism, and it remains unresolved.

Measuring exposure and the reliability problem

Turning the mechanism into a measure requires judging, task by task, what the technology can do, at a scale expert assessment cannot reach. Successive generations of measures bet on different inputs: engineering bottlenecks read at the level of the whole occupation (Frey and Osborne 2017), an approach later criticised for treating occupations as indivisible; the abilities an occupation requires (Felten et al. 2021); the textual overlap of patents with job descriptions (Webb 2019); and the task itself, scored for its suitability for machine learning (Brynjolfsson et al. 2018). Large language models then changed the economics of the measurement. Eloundou et al. (2024) have an LLM rate each O*NET task against a capability rubric, and show the ratings track expert annotators closely while covering the whole taxonomy, which no human exercise can afford. The approach now anchors institutional work: Henseke et al. (2025) build a task-based UK index from LLM ratings, the Office for Budget Responsibility (2025) score every O*NET task to feed a national productivity projection, and the IMF’s complementarity-adjusted index separates displacement from augmentation (Pizzinelli et al. 2023; Cazzaniga et al. 2024).

These measures are not interchangeable. Since they read exposure off different inputs (bottlenecks, abilities, patents, or task text) they rank the same occupation differently, and any estimate built on one inherits that disagreement. The reliability problem has lately sharpened from a cross-measure concern into a within-measure one. Yin et al. (2026) score an identical task set with four contemporary language models and find average exposure differs by more than a factor of three across them, with downstream labour-market magnitudes varying roughly 2.4-fold by model. The divergence is systematic as each model carries a persistent tendency to score high or low, such that averaging does not remove it. That exposure is an indicative quantity, and not the same thing as realised impact, is now firm across the institutional literature. What remains under-developed is any propagation of that measurement uncertainty into the estimates built on top of it.

The geography of exposure and Scotland’s composition

Exposure is a property of tasks and tasks bundle into occupations, so AI’s incidence is uneven across space, though the literature disagrees on how uneven at the spatial scales of policy interest. Sharps et al. (2024) find limited regional variation in UK exposure, attributing it to the diversified sectoral mix of most regions, where the OECD finds cross-country exposure ranging from 16–77% with complementarity rising in regional education (OECD 2024), and European evidence finds automation risk driven systematically by industrial structure and agglomeration (Crowley et al. 2021). The disagreement is partly artefactual: coarse administrative geographies average over the functional labour markets where local conditions actually operate (Overman and Xu 2022), so the regional signal shows up most clearly in occupational mix and least clearly in coarse exposure averages. Consistent with this, Henseke et al. (2025) attribute the recent rise in UK exposure to shifts in occupational composition rather than to within-occupation task change.

Scotland’s occupational mix is distinctive in ways that bear directly on exposure. Its public sector accounts for around 22% of employment against roughly 18% UK-wide, a gap persistent over time and larger than that of any English region, and services account for 77.1% of Scottish GDP against 80.5% for the UK, with more local-government delivery of services such as social care and correspondingly less weight in the professional, scientific and information services concentrated in London and the South East (Government 2024; House of Commons Library 2024). This composition is double-edged. Financial, insurance and professional services are among the most exposed parts of the UK economy, and caring, elementary and skilled-trade work among the least (Office for Budget Responsibility 2025). But the exposed sectors divide on the substitution–complementarity margin, since public administration, health and caring work scores high on complementarity and low on substitutability, whereas professional and financial work is exposed on both. Whether Scotland’s lighter concentration in the latter harbours its workforce or merely denies it the productivity gains of complemented work is therefore ambiguous, and resolvable only empirically. The fiscal edge is sharper still because public-sector pay in Scotland sits about 5% above the UK median and has diverged upward since 2019 (Cribb et al. 2025), against the backdrop of a chronic productivity and business-investment gap (Williams et al. 2025).

Gaps and open questions

The synthesis leaves several gaps. Exposure measures are reported as point estimates whose dependence on the scoring model and prompt is acknowledged but rarely quantified, and none propagates within-model scoring uncertainty into the aggregates built on it. Sub-national exposure for the UK is estimated only at coarse geographies and has not been posed as a compositional question for a devolved fiscal authority, so the Scotland–rUK differential, and which part of it survives the measurement fragility above, is absent from the literature. And no instrument for AI exposure yet matches the commuting-zone design of Acemoglu and Restrepo (2020), so credible sub-national causal estimates are unavailable, and what remains tractable in the near term is the measurement of the exposure differential and the propagation of its uncertainty into forecast implications. These gaps motivate three questions:

1. Which exposure estimands are robust to cross-model divergence? Specifically, do regional differentials survive the model-specific bias that destabilises absolute levels, and can that robustness be established analytically rather than asserted?
2. Does occupational composition insulate or expose a public-sector-intensive economy such as Scotland, once exposure is decomposed on the substitution–complementarity margin rather than read off a single exposure average?
3. How should the measurement uncertainty in LLM-based exposure scores be carried through to the productivity, employment, and wage projections that fiscal bodies such as the SFC and OBR rely on, so that a regional path is reported as a calibrated interval rather than a point?

The remainder of this dissertation takes up these three questions in turn for the Scotland–rUK case.

2 Data

Occupational task content comes from O*NET, regional employment weights and worker-level earnings from the Annual Population Survey (APS), and a published crosswalk links the task data to the survey. I describe each in turn, deferring the construction of the exposure score itself, that is the scoring of tasks, the importance weighting, and the classification rule, to Section 3.

2.1 O*NET task content and the exposure score

O*NET is the occupational database maintained by the US Department of Labor, and is the most comprehensive task taxonomy available for either economy (the United Kingdom has no domestic equivalent).¹ For each O*NET–SOC occupation j the database records a set of task statements describing the discrete activities the work comprises, together with an Importance rating for each task on a five-point scale, measuring the degree to which a specific activity is required for a particular occupation. The task statements are the unit the language model scores, and the importance ratings supply the weights $\omega_{j,t}$ in the occupation-level aggregate of Section 3. The cleaned file contains 17,945 occupation–task pairs across 894 occupations.

The exposure score $E_j \in [0, 100]$ does not appear in the source data, and is therefore a constructed by the procedure set out in Section 3.1. Now, O*NET records the tasks of an occupation, not of an occupation in a place, so a single score attaches to each occupation and carries no regional variation by construction. Additionally, because the scores attach to US occupations, they are of use to a UK study only once mapped onto the domestic classification, which is the role of the crosswalk in Section 2.2.

2.2 Crosswalk to UK SOC 2020

The exposure scores are indexed by O*NET–SOC occupation, whereas the employment weights are recorded under UK SOC 2020. I bridge the two with a published concordance that maps

¹I use release 30.3, available at: <https://www.onetonline.org>

O*NET–SOC 2019 codes to UK SOC 2020 unit groups, from which I form a continuous score E_k^{UK} for minor group k .² I work at the three-digit minor-group level, which trades a degree of occupational resolution for cell sizes large enough to support reliable regional employment shares once survey waves are pooled. That is the binding constraint for Scotland, where finer cells thin out quickly, and the coarser working level is also what makes the region-invariance assumption more comfortable, since averaging over a minor group smooths the fine distinctions along which any residual regional difference in task content would appear (Autor and Handel 2013). The concordance is many-to-many, so an aggregation rule is required; Section B states the rule I use and reports the differential under two alternatives.

2.3 Annual Population Survey employment weights

The APS is the Office for National Statistics’ largest household survey, formed by pooling the quarterly Labour Force Survey with regional sample boosts, and is the standard source for sub-national labour-market statistics. It distinguishes respondents by country and region, allowing Scotland to be observed directly and the rest of the UK (rUK) to be constructed as the aggregate of all other UK regions. From the APS I obtain weighted employment counts by SOC 2020 occupation for each region.

I pool four waves, 2022–2025. Pooling is necessary as a single wave leaves too many Scottish minor-group cells sparsely populated for the employment shares to be precise, and the window is short enough that occupational structure is stable across it. The pooled cell counts $l_{k,r}^{\text{pool}}$ are used to construct the regional employment shares $\eta_{k,r}$ defined in Section 3.2. Together with the crosswalk, they are the only sources of regional variation in the study, as the exposure scores are common to all regions.

The pooled shares remain survey estimates, and Scotland’s thinner cells carry the larger sampling error. This sampling error is independent of the scoring noise of Section 3.3 but of the same order in the differential. I therefore propagate both sources jointly, bootstrapping the pooled shares over persons within region and year and combining the resulting draws with the scoring draws. Table 5 reports each source separately and in combination. The asymmetry arises solely from differences in cell size, and sampling error ultimately accounts for most of the width of the combined interval.

Two features of the APS design qualify the reported interval, and Section B records them without resolving them: the overlapping-wave structure that a person-level bootstrap treats as independent draws, and the 2022–24 fall in LFS response rates, whose differential component would bias the shares rather than widen their interval. Both point the same way, so the composed interval of Table 5 should not be read as an exact coverage statement.

2.4 Earnings

Worker-level earnings come from the same pooled APS waves that supply the employment weights. The dependent variable is the APS derived gross hourly pay for employees, recorded alongside the occupation, industry, region, age, sex, full-time status, highest qualification and public-sector indicator that the wage regression of Section 3.4 uses, so that regression runs on the identical

²See: https://www.nfer.ac.uk/media/1tojrwo0/matching_uk_soc2020_to_o_net.pdf

2022–2025 waves as the employment analysis, weighted by the APS income weight. The earnings and employment results rest on a single consistent source, and, since the APS is public, the wage gradient reproduces without a restricted-data agreement.³

3 Methodology

The strategy has two parts. I build an exposure score for every occupation from language-model ratings of its constituent tasks, then weight those scores by Scottish and rUK employment. The scores are the novel input, so most of this section concerns how they are constructed and how far they can be trusted; the labour-market accounting that follows is comparatively standard.

3.1 Exposure scoring

The OBR score each O*NET task from nought to a hundred for its amenability to AI over a ten-year horizon, separating tasks AI could substitute for from those it would merely complement. They classify an occupation as exposed where enough of its task content clears some threshold, and they map the US result onto UK occupations through a published crosswalk. I adopt that architecture, its scoring scale and its substitute–complement distinction, and depart from it in two ways. Their object is a single national figure, mine the difference between two regions. And they report point estimates without quantifying their dependence on the scoring framework, whereas I treat the scores as model-specific throughout and rest the regional claim on the component of the measurement that survives a change of model.

Procedure. I score the task content of every O*NET–SOC occupation j . For each task $t \in T_j$ the model returns two integer scores on $[0, 100]$: a substitution score $s_{j,t}^{\text{sub}}$, capturing the extent to which AI could perform the task without human involvement over the coming decade, and a complementarity score $s_{j,t}^{\text{comp}}$, capturing the extent to which it could raise a worker’s productivity in the task without replacing them. The prompt fixes the horizon, supplies the task description verbatim from O*NET, and directs the model to the considerations bearing on each judgement, such as whether the task requires physical presence, tacit empathy, or non-delegable accountability.

The two scores must then be combined into a single measure of how exposed a task is. For the exposure measurement I take the larger of the two, $s_{j,t} = \max(s_{j,t}^{\text{sub}}, s_{j,t}^{\text{comp}})$. I choose the maximum for commensurability, since it is the rule behind the OBR’s single-amenability banding. It is not uniquely correct, and it is admittedly coarse. It treats a task scored (90, 10) as no less exposed than one scored (90, 90), and as the maximum of two noisy scores it carries a small upward bias that grows with the framing noise (Clark 1961). Two considerations mitigate this concern. First, the same aggregation rule is applied to each region, so by the logic of Proposition 1 below its effect largely cancels in the regional differential. I verify this by reporting the gap under the substitution score alone and under a saturating alternative, $s_{j,t} = s_{j,t}^{\text{sub}} + (1 - s_{j,t}^{\text{sub}}/100) s_{j,t}^{\text{comp}}$. This alternative treats complementarity as exposing only the share of a task left after substitution,

³The employer-reported Annual Survey of Hours and Earnings (ASHE), securely accessed from the UK Data Service, is the more accurate earnings source, since pay is returned by employers rather than recalled by respondents.

and the resulting differentials remain closely aligned even where the underlying levels differ. Second, the productivity exercise in Section 3.4 does not rely on the collapsed score. Instead, substitution and complementarity enter separately because they have opposite-signed effects in that specification.

Task scores then aggregate to the occupation. The occupation-level score is the importance-weighted mean of its task scores,

$$E_j = \frac{\sum_{t \in T_j} \omega_{j,t} s_{j,t}}{\sum_{t \in T_j} \omega_{j,t}} \in [0, 100], \quad (1)$$

with O*NET importance weights $\omega_{j,t}$, so that an occupation’s exposure reflects its central tasks rather than its peripheral ones. Because the scores attach to US occupations, I map E_j onto a UK SOC 2020 score E_k^{UK} for minor group k through an established crosswalk, working at the three-digit minor-group level so that Scottish employment cells stay populated when survey waves are pooled.

Classification converts the continuous scores into the exposed and unexposed categories. My baseline thresholds are those of the OBR, $(\theta^{\text{sub}}, \theta^{\text{comp}}) = (70, 40)$, applied at the task level: a task is substituted where $s_{j,t}^{\text{sub}} \geq \theta^{\text{sub}}$, complemented where $s_{j,t}^{\text{comp}} \geq \theta^{\text{comp}}$ and the task is not substituted, and unexposed otherwise. The importance-weighted shares of task content in the two exposed classes, Sub_j and Comp_j , then classify the occupation. Occupation j is materially exposed where $\text{Sub}_j + \text{Comp}_j \geq \frac{1}{2}$, and an exposed occupation is substituted or complemented according to the majority of its exposed mass. Where my rule departs is that the OBR band a single amenability score, whereas I elicit the two channels separately and let each face its own threshold. The Scottish composition question turns on the substitution–complementarity margin, and eliciting the two channels separately measures that margin directly, where bands of a single amenability score can only imply it. Sensitivity of the classified indices to the threshold pair is reported at (60, 30) and (80, 50) alongside the baseline.

The resulting score is common to both regions. O*NET records the tasks of an occupation, not of an occupation in a place, so the same E_k^{UK} applies in Scotland and the rUK. To guard against the prompt-sensitivity this literature has repeatedly flagged (Eloundou et al. 2024; Yin et al. 2026), I fix and freeze a single model and prompt before the main run, so that the only variation across occupations is genuine.⁴

3.2 Regional indices and the compositional structure of the gap

The question the Scottish Fiscal Commission needs answered is whether Scotland’s workforce sits more or less in the path of this technology than the rest of the UK’s. The natural summary is an employment-weighted exposure index for each region: how exposed is the average job, given where each region’s employment actually sits across occupations. For region $r \in \{\text{Scot}, \text{rUK}\}$, let $\eta_{k,r} = l_{k,r}^{\text{pool}} / \sum_{k'} l_{k',r}^{\text{pool}}$ be the pooled employment share of minor group k . I form a classified

⁴I record a SHA-256 hash of the frozen system message and user-message template together with every score, so any later change to either is detectable and the figures reproduce from the frozen configuration.

exposed-employment share and a continuous index,

$$\hat{E}_r = \sum_k \eta_{k,r} \mathbf{1}[\text{Class}(k) \in \{\text{Sub}, \text{Comp}\}], \quad \bar{E}_r = \sum_k \eta_{k,r} \frac{E_k^{\text{UK}}}{100} \in [0, 1]. \quad (2)$$

The first answers what fraction of employment sits in materially exposed occupations. The second uses the full scores and is the object for which the formal results below are exact. The gap $\Delta \hat{E} = \hat{E}_{\text{Scot}} - \hat{E}_{\text{rUK}}$, with continuous analogue $\Delta \bar{E}$, is the object of interest.

A shift-share decomposition makes precise what this gap can and cannot contain. Any regional difference in an employment-weighted index has, in principle, two sources: the regions may distribute their employment differently across occupations, or the same occupation may be differently exposed in the two places. Writing the gap as

$$\Delta \hat{E} = \underbrace{\sum_k (\eta_{k,\text{Scot}} - \eta_{k,\text{rUK}}) \mathbf{1}[\text{Exp}(k)]_{\text{rUK}}}_{\text{composition}} + \underbrace{\sum_k \eta_{k,\text{Scot}} (\mathbf{1}[\text{Exp}(k)]_{\text{Scot}} - \mathbf{1}[\text{Exp}(k)]_{\text{rUK}})}_{\text{within-occupation}} \quad (3)$$

separates the two sources. Since the same score applies on both sides of the border, the within-occupation term vanishes identically. The entire gap is therefore compositional. Any Scotland–rUK differential reflects Scotland’s distinctive occupational mix, with more public administration and caring work and less professional and financial services.

This compositional reading rests on one assumption: the within-occupation task profile of each UK minor group is the same in both regions, so that a single E_k^{UK} describes minor group k on either side of the border. The assumption is not that occupations are internally homogeneous; Henseke et al. (2025) show they are not, with the highest score dispersion in cells where a generic task description admits both an exposed and an unexposed reading. It is the weaker claim that whatever heterogeneity exists is distributed similarly in the two regions.

Assumption 1 (Region invariance). Writing T for a worker’s task bundle, k for their minor group and $R \in \{\text{Scot}, \text{rUK}\}$ for region, the conditional distribution of tasks is region-invariant,

$$\Pr(T \in A \mid k, R = \text{Scot}) = \Pr(T \in A \mid k, R = \text{rUK}) \quad \text{for all } A. \quad (4)$$

Internal homogeneity would require T to be degenerate given k . What I need is only Equation (4), so that whatever within-occupation heterogeneity exists is distributed identically on either side of the border. Even this overstates what the differential requires: writing $e(\cdot)$ for task-level exposure, only the conditional means need agree, $\mathbb{E}[e(T) \mid k, \text{Scot}] = \mathbb{E}[e(T) \mid k, \text{rUK}]$, and any common departure of that mean from the O*NET-measured profile cancels from $\Delta \hat{E}$ in any case. Within one national economy this is reasonable. The qualifications, technologies, and regulatory standards that define an occupation are set at UK level, and the coarse working level of the crosswalk averages over the fine distinctions along which any residual regional difference would appear (Autor and Handel 2013). The assumption fails only if task content differs across regions in a way correlated with exposure. I cannot test this directly without region-specific task data, but the international evidence suggests such within-occupation differences are second-order relative to composition (Handel 2012; Autor and Handel 2013).

Finally, a single E_k^{UK} attaches to a minor group irrespective of the industry in which the occupation is employed. As a result, any occupation-by-industry variation in exposure, between the same accountant working in a bank and in a local authority, say, is removed from the regional differential by construction. The design can still identify whether the Scotland–rUK gap arises from differences in industry composition or from differences in occupational composition within industries, and Section 4.3 decomposes the differential along those lines. What it cannot capture is variation in exposure within a given occupation–industry cell across regions. Doing so would require industry-specific task data, which no available source provides.

3.3 Scoring error and the robustness of the differential

A task score can be wrong for more than one reason, and the design responds to each of them separately. Meaning-preserving changes to the prompt move the score, and that framing noise is what the calibration of Equation (9) measures. Each model carries a persistent tendency to over- or under-score, and that systematic bias (Yin et al. 2026) is what Proposition 1 addresses and the cross-model test of Section 4.7 probes. Aggregation and crosswalk error is bounded by the operator and threshold robustness, and sampling error in the employment weights by the APS bootstrap; both are handled where they arise.

The cross-model divergence documented by Yin et al. (2026) would be fatal to a study reporting absolute levels as objective measurements. It is far less threatening to a study of regional differences, however, and this subsection formalises why. Consider a score generated by model- M , decomposed as

$$s_j = \theta_j^* + b_j^M + \varepsilon_j, \quad \varepsilon_j \sim N(0, \sigma_{g(j)}^2), \quad (5)$$

where θ_j^* is the unobserved cross-model consensus, b_j^M is model M 's systematic bias, meaning its persistent tendency to over or under-score occupation j , and ε_j is mean-zero framing noise indexed by the occupational family $g(j)$ defined in the calibration below. The consensus θ_j^* is a device for organising the decomposition, not an estimand: only contrasts between models are ever observed, and nothing below requires access to it. The two error terms play different roles and therefore require different treatment. The noise term ε_j is the component that averaging and simulation can reduce. b_j^M , by contrast, is the persistent model-specific bias identified by Yin et al. (2026), affecting every absolute estimate a model produces. I measure the former directly and show that the latter cancels in the regional differential.

Conditioning on the frozen model absorbs b_j^M into a model-specific target $\theta_j^M = \theta_j^* + b_j^M$, leaving the classical measurement model $s_j = \theta_j^M + \varepsilon_j$, with group variances $\sigma_{g(j)}^2$ to be calibrated below. Writing the continuous UK score as $E_k^M = E_k^* + b_k^M$, the cancellation result follows.

Proposition 1 (Common-mode cancellation). *The model-induced error in the differential $\Delta \bar{E}^M = \bar{E}_{\text{Scot}}^M - \bar{E}_{\text{rUK}}^M$ is*

$$\Delta \bar{E}^M - \Delta \bar{E}^* = \frac{1}{100} \sum_k (\eta_{k,\text{Scot}} - \eta_{k,\text{rUK}}) b_k^M. \quad (6)$$

Hence (i) a pure level shift $b_k^M = b^M$ leaves the differential exact, $\Delta \bar{E}^M = \Delta \bar{E}^*$; and (ii) in general only the part of the bias co-varying with the regional employment-share gap survives, with

the mean bias being removed by construction.

Section A proves Proposition 1 and states the demeaning step it turns on.

The absolute index $\bar{E}_r^M$ carries contamination of the order of the mean bias, which is what the three-fold divergence of Yin et al. (2026) measures. The differential retains only the part of that bias co-varying with the cross-region employment-share gap. So the same compositional weighting that zeroes the within-occupation term of Equation (3) also removes the common-mode bias, and the differential is robust where the levels are fragile. How much of the bias in fact survives is an empirical matter, and Section 4.7 measures it.

The proposition also says something a re-scoring experiment can check, and I record that implication as a corollary so the cross-model test of Section 4.7 has a formal target to be judged against.

Corollary 1 (Cross-model stability). *For any two models M and M' ,*

$$\Delta \bar{E}^M - \Delta \bar{E}^{M'} = \frac{1}{100} \sum_k \Delta \eta_k (b_k^M - b_k^{M'}), \quad |\Delta \bar{E}^M - \Delta \bar{E}^{M'}| \leq \frac{1}{100} \|\Delta \eta\|_2 \|b^M - b^{M'}\|_2, \quad (7)$$

where $\Delta \eta_k = \eta_{k,\text{Scot}} - \eta_{k,\text{rUK}}$ and the bias differences are demeaned. The cross-model movement in the differential is zero under a common level shift and bounded by the alignment of the bias difference with the share gap in general.

Equation (5) is additive, so Proposition 1 protects the differential against an additive bias only. A model may instead disagree about the *scale* of the index, compressing or stretching the score distribution while preserving the ordering. The next corollary covers that case, which Section 4.7 shows is the one the data present.

Corollary 2 (Affine bias and scale-free differentials). *Suppose model M 's scores are affine in the consensus up to an idiosyncratic term, $E_k^M = a^M + c^M E_k^\star + b_k^M$ with $c^M > 0$. Then*

$$\Delta \bar{E}^M = c^M \Delta \bar{E}^\star + \frac{1}{100} \sum_k \Delta \eta_k b_k^M. \quad (8)$$

Hence (i) the intercept a^M cancels exactly, by Equation (15), and the slope rescales the differential proportionally rather than cancelling; (ii) writing SD^M for the employment-weighted standard deviation of $\{E_k^M\}$, the standardised differential $\Delta \bar{E}^M / \text{SD}^M$ removes both a^M and c^M , exactly when $b_k^M \equiv 0$ and up to the residual term otherwise; and (iii) replacing each score by its employment-weighted percentile rank yields a differential invariant to any strictly increasing transformation of the scale, of which the affine map is a special case.

A statement about the differential is *scale-free* when it does not depend on how widely the index happens to be spread under a given model. Such a statement locates Scotland relative to the dispersion of UK occupations, and not at a distance in one model's units. Corollary 2 offers two such statements because they buy different things. Dividing by the model's own dispersion keeps the differential cardinal, so it remains comparable with other standardised gaps, but undoes only an affine distortion. Ranks undo any monotone distortion whatever, at the price of discarding everything but order.

The results answer different failures. Proposition 1 handles a model that reads the whole economy as more or less exposed, Corollary 2 one that uses more or less of the available scale, and the percentile form one that agrees only about the ordering, the last being non-parametric and correspondingly the weakest statement about magnitude. That is, re-drawing the scores from a different model should move the levels, rescale the differential, and leave its sign and scale-free magnitude where they stood. I test exactly this by re-scoring the full task set on two alternative models and comparing the cross-model stability of $\Delta \bar{E}$ against that of $\hat{E}_{\text{Scot}}$ and $\hat{E}_{\text{rUK}}$ individually.⁵ If the levels move while the gap holds, the regional finding survives the critique that undermines the absolute numbers.

Calibration and propagation. The differencing handles the bias but the noise has to be measured. Framing noise is the variation in a score under changes to the prompt that leave its meaning intact. I do not assume a functional form for it. I estimate its variance from a stratified experiment. The experiment draws $N = 90$ occupations, eighteen from each of the five broad occupational families obtained by collapsing the O*NET major groups onto their leading digit: ten selected at random to estimate the variance and eight nearest the classification boundary, held out for the stability check. The five-way grouping is coarse deliberately. It is the finest partition at which every cell carries enough occupations for a variance estimated from four scoring conditions to be usable, and $g(j)$ denotes it throughout. Each sampled occupation’s full task set is re-scored under the conditions of Table 1. Four of these conditions (the baseline, the assessment criteria in reverse order, a paraphrased rubric of identical content, and a reordered task presentation) are meaning-preserving perturbations of the frozen prompt, and the disagreement among them is what identifies the noise. Writing K_w for this set, with $|K_w| = 4$,

$$\hat{\sigma}_j^2 = \frac{1}{|K_w| - 1} \sum_{k \in K_w} (s_j^{(k)} - \bar{s}_j)^2, \quad \hat{\sigma}_g^2 = \frac{1}{|J_g|} \sum_{j \in J_g} \hat{\sigma}_j^2, \quad (9)$$

where J_g collects the random-stratum occupations of family g . Two further conditions, a five-year horizon and a single composite score, change the question rather than its phrasing, and a shorter horizon is not an error in measuring the ten-year one. I therefore report their occupation-level shifts as named sensitivities and do not pool them into $\hat{\sigma}_g^2$.⁶

With the variance in hand, I propagate it into the continuous aggregates by simulation. Across $R = 1,000$ draws I perturb each occupation score, $s_j^{(r)} \sim N(s_j, \hat{\sigma}_{g(j)}^2)$ truncated to $[0, 100]$, re-aggregate to UK minor groups, re-weight, and record $\Delta \bar{E}^{(r)}$. I report the resulting 95% interval and the share of draws in which the gap keeps its sign.⁷ The uncertainty propagation is carried out on the continuous index by design. It is the object for which Proposition 1 holds exactly and

⁵I use one model from a different provider (OpenAI’s GPT-4o) and one from a different tier of the same provider (Anthropic’s Claude Haiku 4.5).

⁶Pooling them would conflate framing noise with a change of estimand and overstate the variance to which the propagation applies. By Proposition 1, the component of either shift common across occupations is in any case innocuous for the differential.

⁷A gap whose interval excludes zero is robust to scoring noise, and one whose interval includes zero is not. As a check I allow intra-group correlated errors, $\varepsilon_g^{(r)} \sim N(0, \hat{\sigma}_g^2[(1 - \rho)I + \rho \mathbf{1}\mathbf{1}^\top])$ for $\rho \in \{0, 0.2, 0.4\}$, where ρ is the degree to which scoring errors are assumed to be correlated within an occupational family, so the higher values are the cautious cases. Correlated framing errors would otherwise lead independent draws to understate aggregate uncertainty. Section A derives the resulting inflation factor.

Table 1: Calibration design: scoring conditions and the role of each

Variant	What changes	Role
V0	Baseline prompt, 10-year horizon	noise (reference)
V1	Assessment criteria in reverse order	noise: order and anchoring
V2	Paraphrased rubric, identical content	noise: wording
V3	Task presented before occupation	noise: framing
V4	5-year horizon, else identical	estimand sensitivity: horizon
V5	Single composite score	estimand sensitivity: rubric form

The within-model variance of Equation (9) is estimated from V0, V1, V2 and V3 on the random sample only; V4 and V5 are reported as named sensitivities. Cross-model contrasts come from the full re-scores under the two alternative models, not from a calibration condition.

for which the noise model in Equation (9) is defined. Alongside the scoring draws, I bootstrap the pooled employment shares over persons within region and year and combine the two independent sources of uncertainty to construct an interval for $\Delta\bar{E}$. To show their relative importance, I also report the scoring and sampling intervals separately.

The classified share needs a separate treatment, because it is nonlinear at the thresholds. A small perturbation to a task sitting near a cut-off can flip its class, and with it the classification of an occupation. I therefore give each occupation a stability score: the importance-weighted probability that its task content keeps its class under re-framing, obtained by reading each task’s distance from the nearest cut-off in units of the calibrated noise standard deviation and passing that distance through the normal cdf. Section D states the construction and the reporting landmark I flag against. In the event the flag binds nowhere. No occupation and no minor group falls below the landmark, so Section 4.5 compares the held-out boundary stratum against the random one instead of comparing flagged occupations against unflagged ones, and the stability apparatus enters as a null diagnostic.

3.4 Empirical specifications

With exposure scores and regional weights in hand, I put the measured differential to work on three objects the Scottish Fiscal Commission forecasts, carrying the scoring uncertainty of Section 3.3 through into each. The productivity projection is the one strictly necessary to the fiscal question, since it alone converts an exposure gap into the units in which a block-grant-adjustment claim would have to be made. The employment gradient restates the compositional reading of Equation (3) in observable quantities, and the wage gradient is the micro counterpart of the aggregate propagation, where the calibrated scoring variance meets a regressor measured with error. None is offered as a causal estimate.

The first step expresses the compositional gap as an estimable gradient, regressing log occupational employment, and in a second specification the employment share, on the continuous exposure score, a Scotland indicator and their interaction, within one-digit SOC major groups and weighted by cell employment. The interaction asks whether, within broad groups, Scottish employment loads differentially onto exposed work: a negative estimate would mean composition insulates Scotland, a positive one the reverse. Section F gives the specification in full and Section 4.3 reports the estimates, which restate the decomposition of Equation (3) in observable quantities and add nothing to it.

The second step translates the exposure differential into productivity, using the task-based framework of Acemoglu (2025) as parameterised by the Office for Budget Responsibility (2025). In that framework, AI raises total factor productivity by lowering the cost of the tasks it can perform, so the gain to a region depends on how much of its employment sits in exposed work. The proportional TFP gain for region r is

$$\Delta \ln \text{TFP}_r \approx \kappa \sum_k \eta_{k,r} \pi_k \bar{E}_k, \quad (10)$$

with π_k the task-cost share and κ the calibrated average cost saving on exposed tasks. Substitution and complementarity carry different economics here. The first is a cost saving on tasks the technology takes over; the second augments the worker who is retained. I therefore run the projection on the two channels separately rather than on their maximum, mapping the substitution channel to the cost-saving term above and reporting the complementarity channel alongside it. I also report the composite version for comparability with the OBR exercise. I import κ and the horizon parameters from the OBR/Acemoglu calibration rather than estimating them, so the Scottish and rUK paths differ only through their employment-weighted exposure. Acemoglu (2025) builds κ from the two task-level randomised experiments of Noy and Zhang (2023) and Brynjolfsson et al. (2025), averaging their labour cost savings of 40% and 14% to about 27%, then converting to overall cost savings using industry labour shares; the baseline calibration sets the task-cost weight to $\pi_k = \pi = 0.23$, his share of exposed tasks judged profitably automatable over the decade, uniform across occupations. Because π_k is constant in k , the regional differential in each channel’s implied TFP path inherits the cancellation of Proposition 1 exactly, by Corollary 3; only the levels of the path carry the model dependence. Section F derives Equation (10) and gives the calibration in full, and I report the implied output-per-worker gap as a band reflecting the OBR’s own parameter range.⁸

The third step estimates the exposure–wage relationship at worker level,

$$\ln w_{i,t} = \alpha + \beta_1 E_{j(i)} + \beta_2 \text{Scot}_i + \beta_3 (E_{j(i)} \times \text{Scot}_i) + \gamma X_{i,t} + \delta_s + \delta_t + \varepsilon_{i,t}, \quad (11)$$

where $\ln w_{i,t}$ is the log gross hourly pay of worker i in survey year t , $E_{j(i)} \in [0, 1]$ is the continuous exposure score of the worker’s minor group $j(i)$, common to both regions and constant within an occupation, and Scot_i indicates residence in Scotland. The coefficient β_1 is the rUK gradient of pay in exposure, β_2 the Scotland wage-level shift, and β_3 , the object of interest, the Scotland-specific rotation of that gradient. The controls $X_{i,t}$ are age, its square, sex and full-time status; δ_s and δ_t are industry-section and survey-year fixed effects; and observations are weighted by the APS income weight. Two further controls enter the richer specifications of Table 9. The first is highest qualification, the classic omission of the earnings-function literature: exposed occupations are graduate-heavy, so any Scotland–rUK difference in the within-occupation qualification profile would otherwise load onto β_3 . The second is a public-sector indicator, because I read the estimated rotation partly through the public-sector pay premium (Cribb et al. 2025) and Scotland’s public sector is proportionally larger, so I test that reading against the indic-

⁸The contribution is the regional differential in the implied path, not a reassessment of Acemoglu’s headline magnitude, which is contested and outside the scope of this dissertation.

ator. $E_{j(i)}$ varies only across occupations, every worker in a minor group shares one value of the regressor and the within-group correlation of residuals inflates conventional standard errors by the Moulton factor (Moulton 1990). I therefore cluster by occupation, the level at which the regressor varies.

4 Results

I first establish the measure on which the comparison rests, then present and decompose the Scotland–rUK differential. I next examine its sensitivity to the measurement choices that, according to Yin et al. (2026), undermine the interpretation of absolute exposure levels. The differential is then translated into the quantities the SFC forecasts, with the scoring uncertainty propagated through each stage.

4.1 Occupational exposure and the saturation of the classified measure

The binary exposed/unexposed measure that dominates the policy literature cannot carry the regional question, because it is saturated. The baseline run scores all 17,945 task–occupation pairs across 894 O*NET occupations. Aggregated to the 104 UK SOC 2020 minor groups on which the regional comparison rests, E_k ranges from 31.7 (building finishing trades) to 84.3 index points (business, research and administrative professionals), with a cross-group standard deviation of 13.6 points, so the continuous measure carries the between-occupation variation a compositional comparison needs. The employment-weighted means, 66.7 points in rUK and 65.8 in Scotland, are high by the standards of the survey-based literature, and since the level is the object Section 3 argues is fragile to the scoring model, I place no weight on them.

The classified measure preserves little of this variation. Under the central thresholds, 84 of the 104 minor groups are Complemented, 9 Substituted, and 11 Unexposed, so that 90.8% of rUK employment and 90.9% of Scottish employment sits above the materiality cut. The classifier is, in effect, right-censored: once roughly nine-tenths of employment in both regions is coded as exposed, any regional difference in the intensity of exposure is lost. The classified gap is, then, driven largely by composition within the small unexposed tail. For this reason, I treat the continuous index as the primary measure and use the classified shares mainly as a diagnostic of the classification scheme itself.

4.2 The Scotland–rUK differential

Table 2 contains the central result. Scotland’s employment-weighted continuous exposure is 0.86 index points below rUK on the 0–100 scale (65.82 against 66.69), a relative shortfall of 1.3%. The sign is invariant to the exposure operator: the substitution-only index gives -0.35 index points, the complementarity index -0.96 , and the saturating combination -0.77 . The composed 95% interval of Section 4.6, carrying scoring noise and APS sampling error jointly, is $[-1.07, -0.69]$ index points. Scotland is therefore less exposed than the rest of the UK, precisely enough to rule out zero, though by a modest margin.

The durable form of that result is scale-free. Scotland sits about two percentile points below rUK in the UK occupational exposure distribution, a magnitude Section 4.7 shows to be stable

Table 2: Regional exposure indices and the Scotland–rUK differential

	Scotland	rUK	Gap	Relative gap
<i>Panel A: continuous indices (index points, 0–100)</i>				
Composite (max operator)	65.82	66.69	−0.86	−1.3%
Substitution only	38.30	38.65	−0.35	−0.9%
Complementarity	62.70	63.66	−0.96	−1.5%
Saturating combination	74.74	75.51	−0.77	−1.0%
<i>Panel B: classified employment shares (%)</i>				
Exposed	90.91	90.78	+0.14	+0.1%
Substituted	14.90	14.21	+0.68	+4.8%
Complemented	76.02	76.56	−0.55	−0.7%

Notes: Panel A is the continuous exposure index on a 0–100 scale and its gaps are in index points; Panel B reports shares of regional employment and its gaps are in percentage points of employment, under the central thresholds (substitution ≥ 70 , complementarity ≥ 40).

across scoring models where the -0.86 index-point figure is not, and it is the percentile statement a forecasting body should carry forward. The index-point gap is nonetheless reported throughout, since the decomposition, the propagation and the productivity functional are all defined on that scale, and should be read as the baseline model’s expression of a finding whose robust content is ordinal.

The differential is small because the two regional labour markets are similar, which is a feature of the data and not of the measurement. Their employment-share vectors are close to collinear, with the largest difference across the 104 minor groups amounting to only 1.6 percentage points of employment. Over the observed score range of 32–84 index points, no assignment of exposure can produce a large aggregate gap from compositional differences this small. What the continuous index adds is the ability to recover the sign of that gap and trace it to its underlying components.

4.3 Anatomy of the differential

The gap decomposes exactly into per-occupation contributions $\Delta\eta_k(E_k - \bar{E}_{\text{rUK}})$, and Table 3 reports the largest on each side. The negative contributions come from two sources. Scotland is underrepresented in highly exposed professional occupations (functional managers and directors alone contribute -0.17 points and IT professionals -0.11) while being overrepresented in mid-exposure service occupations that a binary classifier would place in the same category as the top of the distribution, notably caring personal services (-0.17) and other elementary services (-0.16). The offsetting positive contributions arise differently. Scotland is underweight in some low-exposure work, chiefly childcare and related personal services ($+0.09$) and construction and building trades ($+0.06$), and it is overweight in a few high-exposure occupations, among them engineering professionals ($+0.05$), government administration ($+0.05$) and customer service ($+0.04$), the last two being minor groups in which substitution dominates the exposure mix. The differential emerges from many small and economically coherent compositional differences rather than from a few influential occupations.

Nor is the differential simply a London effect. London’s index is 69.30, compared with 66.33 for the UK excluding London, while Scotland’s 65.82 remains 0.51 index points below the ex-London comparator. Excluding London therefore closes 41% of the headline gap, but it does

Table 3: Decomposition of the continuous differential: largest contributions

SOC	Minor group	E_k	$\Delta\eta_k$ (pp)	Contribution (pp)
113	Functional managers and directors	77.5	−1.58	−0.171
613	Caring personal services	51.2	+1.07	−0.165
926	Other elementary services	46.3	+0.76	−0.155
922	Elementary cleaning	39.1	+0.51	−0.140
213	IT professionals	80.9	−0.79	−0.112
611	Childcare and related personal services	47.5	−0.49	+0.094
531	Construction and building trades	37.4	−0.20	+0.059
212	Engineering professionals	79.3	+0.42	+0.053
411	Administrative: government and related	79.4	+0.39	+0.050
721	Customer service occupations	79.4	+0.35	+0.044

Notes: Contributions are $\Delta\eta_k(E_k - \bar{E}_{rUK})$, in index points, summing exactly to the composite gap of −0.86 index points; the five largest on each side are shown. E_k is on the 0–100 exposure scale and $\Delta\eta_k$ is the Scotland-minus-rUK difference in the within-region employment share, in percentage points of employment.

Table 4: Industry-mix and within-industry components of the differential

Term	Ordering A (rUK exposure, Scottish shares)	Ordering B (Scottish exposure, rUK shares)
Industry mix	−0.2094	−0.1708
Within industry	−0.6561	−0.6947
Total (= gap)	−0.8655	−0.8655

Notes: Index points on the 0–100 exposure scale. Employment is split by SIC 2007 section on the pooled APS 2022–2025 sample for which industry is recorded and the occupation matches a scored minor group, so the total differs slightly from the headline gap of Table 2. Per-section contributions are in Table 12.

not eliminate it. The professional-management underweight persists (functional managers still contribute −0.14 points) as does the relative concentration in care-sector employment. Scotland ranks fifth among the twelve UK nations and regions, behind London, the South East, the East of England, and the North West. The appropriate conclusion is not that Scotland is unusually insulated from AI exposure. Rather, London is unusually exposed, while Scotland sits slightly below the middle of a distribution that spans barely 2.5 index points once London is excluded.

The employment regression in Section F agrees that within broad occupation groups, Scottish employment sits systematically further down the exposure gradient than rUK employment, on the log-employment and employment-share margins alike.

The industry decomposition points to the same conclusion. The industry-mix component accounts for only −0.17 to −0.21 of the −0.87 index-point gap, while the within-industry component accounts for −0.66 to −0.69 (Table 4). Roughly three-quarters to four-fifths of the differential therefore arises within industries, and the within-industry term is negative in 17 of the 21 SIC sections (Table 12), with education the only material exception. Even holding the industry distribution fixed, Scottish employment within industries sits lower on the exposure gradient.

By contrast, the industry-mix component is small because sectoral differences net out. Scotland is underweight in professional, scientific and information services but overweight in public administration, mining and energy, and marginally overweight in finance. The gap is therefore a professional-services phenomenon rather than a financial one.

4.4 Substitution against complementarity

The channel through which Scotland is exposed also differs from rUK, and the distinction matters because substitution and complementarity carry opposite-signed productivity implications in the projection of Section 4.8. Panel B of Table 2 shows Scotland’s classified substituted employment share 0.68 percentage points above rUK (14.90% against 14.21%), even though its continuous substitution intensity is 0.35 points lower. Scotland is overweight in occupations where the substitution score exceeds the complementarity score, such as sales assistants and customer-service occupations, whilst underweight in the administrative minor groups with the highest substitution intensities. The precise magnitude depends on how exposure is classified, but the qualitative pattern is stable across specifications. That is, Scottish exposure is relatively more concentrated in substitution-dominated occupations, even though both continuous channel indices are lower than in rUK. The continuous indices make the same point without the classifier: Scotland’s substitution deficit is proportionally smaller than its complementarity deficit (-0.9% against -1.5%), so its exposure leans towards substitution in relative terms even while sitting below rUK on both channels. For that reason, the productivity projection in Section 4.8 treats the two channels separately throughout.

4.5 Fragility of the classified gap, robustness of the continuous gap

The classified gap changes sign under reasonable variations in its own construction whereas the continuous gap does not. At the central thresholds Scotland’s exposed share exceeds rUK’s by 0.14 percentage points, at the low thresholds by 0.19 points, and at the high thresholds the sign reverses to -1.36 points. A measure whose sign depends on the threshold pair is a diagnostic of its own construction, and the mechanism is the censoring of Section 4.1: with 91% of employment above the cut, the classified gap is generated by a handful of share differences in the unexposed tail, and moving the cut relocates the tail. The continuous gap, by contrast, is negative under all four operators, under both classification schemes’ underlying scores, and at every point of the uncertainty analysis below.

Classification stability behaves consistently with this reading. Across the meaning-preserving variants, 55.0% of the near-boundary calibration occupations flip class against 6.0% of the randomly drawn ones, so instability concentrates where it should. It does not propagate: no occupation and no minor group falls below the stability landmark of Section D, so the flag never binds, and excluding minor groups by average stability leaves the classified gap unchanged to the fourth decimal place at every cutoff up to 0.85. The flipping occupations carry small and regionally similar employment shares.

4.6 Scoring noise, estimand sensitivity, and the composed interval

The calibration experiment delivers the within-model noise parameters. Across the meaning-preserving variants (V1–V3 of Table 1), the standard deviation of an occupation composite ranges from 1.3 to 2.4 index points across the five occupational families, with task-level standard deviations of 3.4–5.1 index points for substitution and 3.6–6.0 for complementarity. The estimand variants shift levels by much more. Shortening the horizon to five years (V4) moves composites

Table 5: Propagated uncertainty in the differential $\Delta \bar{E}$

Component	ρ	Mean	SD	2.5%	97.5%
Scoring only	0	-0.863	0.017	-0.897	-0.830
Scoring only	0.2	-0.864	0.031	-0.922	-0.801
Scoring only	0.4	-0.864	0.039	-0.938	-0.788
Sampling only	-	-0.864	0.099	-1.070	-0.690
Composed	0	-0.863	0.101	-1.069	-0.688
Composed	0.2	-0.862	0.103	-1.086	-0.677
Composed	0.4	-0.865	0.108	-1.091	-0.675

Notes: Index points on the 0–100 exposure scale; 1,000 draws per row. Scoring draws perturb occupation composites with the calibrated σ_g , correlated within occupational family at ρ ; sampling draws bootstrap the employment shares over persons within region and year. The scoring perturbation is common to both regions by construction, so its rows illustrate Proposition 1 rather than quantify uncertainty in the gap.

by -1.5 index points on average, and collapsing the two-dimensional rubric to a single score (V5) moves them by -11.6 . Yet the cross-occupation correlations with the V0 baseline remain 0.998 and 0.956, so changing the question moves every level and barely moves the ordering. That is the same common-mode structure Proposition 1 formalises, with levels highly specification-dependent and relative positions not.

Table 5 propagates the noise, though the scoring-only rows are best viewed as a diagnostic. Each draw perturbs the occupation scores once and feeds both regions the same perturbed vector, so scoring noise is common-mode across regions by construction, and its near-total cancellation in the differential (an SD of 0.017 index points against level SDs fifty times larger) is a numerical illustration of Proposition 1, not a credible interval for the gap. The relevant uncertainty comes from sampling. Bootstrapping the employment shares widens the SD to 0.099 index points, and the composed interval at the most conservative within-family correlation is $[-1.09, -0.68]$ index points, with all 1,000 composed draws negative. Together this frames the headline exposure result as $\Delta \bar{E} = -0.86$ index points, with 95% interval $[-1.07, -0.69]$, most of whose width comes from survey sampling.

4.7 Cross-model stability of the differential

The remaining concern is model choice. As Yin et al. (2026) show, it introduces a source of error that averaging cannot remove and therefore provides a direct test of Proposition 1. I re-scored all task–occupation pairs on two additional models, GPT-4o (a different provider) and Claude Haiku 4.5 (a lower tier of the same provider), and recomputed the differential from each.

The levels behave as Yin et al. (2026) would predict. The alternative models disagree with the Sonnet baseline about absolute exposure by a mean occupation-level bias of 3.5 index points for Haiku and 9.1 for GPT-4o, the latter a divergence of the order Yin documents across frontier models. The differential moves far less. Against a baseline gap of -0.86 index points, Haiku returns -0.53 and GPT-4o -0.43 , so the differencing removes on the order of nine-tenths of the model disagreement. That much is Proposition 1 doing its work on the additive component of the bias.

Corollary 2 shows that the cross-model differences are systematic rather than idiosyncratic. In both alternative models the bias is strongly correlated with the exposure level itself (0.88 for

Table 6: Cross-model stability of the differential

Scoring model	Contrast	Mean bias	$\Delta \bar{E}$	$\text{corr}(b_k^M, \Delta \eta_k)$
Claude Sonnet 4.6	baseline	—	−0.86	—
Claude Haiku 4.5	same provider, lower tier	3.5	−0.53	−0.24
GPT-4o	different provider	9.1	−0.43	−0.23

Notes: Index points of the 0–100 scale. “Mean bias” is the mean absolute occupation-level score difference from the Sonnet baseline, $|b_k^M|$; $\Delta \bar{E}$ is the employment-weighted Scotland–rUK differential recomputed from each model’s scores; b_k^M is the minor-group score difference from baseline, in index points, and $\Delta \eta_k$ the Scotland–rUK employment-share gap, so the final column is the empirical analogue of the surviving term in Equation (6).

Table 7: Scale-free cross-model differentials

Model	Score SD (index pts)	Raw gap (index pts)	SD-unit gap (SDs)	Percentile gap (pctile pts)
Claude Sonnet 4.6 (baseline)	13.582	−0.863	−0.0636	−1.966
Claude Haiku 4.5	9.593	−0.531	−0.0554	−1.756
GPT-4o (2024-11-20)	7.549	−0.431	−0.0571	−2.008

Notes: The score SD is the UK-employment-weighted standard deviation of the 104 minor-group scores under each model. The SD-unit gap divides the index-point gap by that dispersion, removing any affine rescaling; the percentile gap replaces each score with its UK-employment-weighted percentile rank, removing any monotone rescaling. Both transformations are applied to the score distribution, which is common to the two regions, so neither introduces regional variation. Model biases and the raw differentials are in Table 6.

Haiku, 0.95 for GPT-4o), implying a compression of the scale rather than a simple shift.⁹ The employment-weighted standard deviation of the minor-group scores falls from 13.58 index points under Sonnet to 9.59 under Haiku and 7.55 under GPT-4o, implying affine slopes of $c^M = 0.71$ and 0.56. Equation (8) then predicts differentials of −0.61 and −0.48 from compression alone, against realised values of −0.53 and −0.43, so the affine term accounts for roughly three-quarters of the movement in the Haiku gap and nine-tenths in the GPT-4o gap. Because compression pulls the highest-exposure occupations down and Scotland is underweight in exactly those, it correlates mildly with the share gap (−0.24 and −0.23), which is how a multiplicative disagreement that Equation (5) cannot represent presents itself in Equation (6).

Parts (ii) and (iii) of Corollary 2 say what to do about it. Dividing by each model’s score dispersion removes any affine rescaling and percentile ranks remove any monotone one, and under these transformations the raw gaps of −0.86, −0.53 and −0.43 index points become −0.064, −0.055 and −0.057 standard-deviation units and −1.97, −1.76 and −2.01 percentile points (Table 7). The two-fold disagreement in index points collapses to about 13%, and GPT-4o, whose raw gap lies outside the within-model interval of Table 5, lands within 2% of the baseline in percentile terms. Those standardised gaps are the non-affine residual of Equation (20), so their spread measures how much of the disagreement concerns Scotland’s position within the distribution, and on that the three models are close.

This narrows the claim and improves its foundation. The sign of the differential, its compositional structure, and its scale-free magnitude of about two percentile points all survive a change

⁹This is a non-classical, mean-reverting error of the kind validation studies of survey data routinely find (Bound and Krueger 1991), with the difference that here there is no validated truth to revert towards, only a baseline that is itself a model.

of scoring model, and Corollary 2 guarantees the last of these against any monotone rescaling instead of leaving it as an observed regularity. The index-point figure does not survive. It ranges from -0.43 to -0.86 because the models disagree about the dispersion of the index far more than about Scotland’s position within it. I therefore report the scale-free magnitude as the headline, with -0.86 index points as the baseline model’s expression of it.

4.8 Implications for projected productivity

The projection converts the exposure gap into the units a fiscal claim has to be made in. Since the baseline projection is a linear functional of the scores with region-common coefficients, Corollary 3 makes its regional differential inherit the cancellation of Proposition 1 exactly, so the differential is proportional to $\Delta \bar{E}$ by construction.

Under the central adoption scenario the composite index implies a cumulative ten-year TFP contribution of 2.209% for rUK against 2.180% for Scotland, a differential of -0.029 percentage points (Table 8). Moving the average cost saving on an exposed task, κ , across its low and high settings scales the rUK level from 1.69% to 2.79% and the differential with it, from -0.022 to -0.036 points. Run on the two channel indices separately rather than on their composite, the complementarity projection gives -0.032 percentage points and the substitution projection -0.012 . These are alternative projections of the same gap rather than additive components of the composite figure, and in this case, fall either side of it.

Now, equation (10) asks for the share of an occupation’s task content that is exposed, and I use the continuous index $\bar{E}_k$ in its place. The two are not the same object: $\bar{E}_k$ is an importance-weighted mean amenability rating, whereas the framework wants an importance-weighted *share* of task content above a cut. Panel C therefore repeats the central projection on the latter, $\text{Sub}_k + \text{Comp}_k$, at each of the three threshold pairs. Now, equation (10) asks for the share of an occupation’s task content that is exposed, and I use the continuous index $\bar{E}_k$ in its place. $\bar{E}_k$ is an importance-weighted mean amenability rating, whereas the framework wants an importance-weighted *share* of task content above a cut. Panel C therefore repeats the central projection on the latter, $\text{Sub}_k + \text{Comp}_k$, at each of the three threshold pairs. The differential holds its sign throughout, ranging from -0.021 to -0.050 points meaning whichever functional stands in for Acemoglu’s exposed task share, Scotland’s projected productivity gain from AI adoption falls short of rUK’s.

Neither functional licenses a reading of the levels. Both put a far larger fraction of task content in the exposed category than the roughly 20% Acemoglu takes from Eloundou et al. (2024), so both inflate the projected path relative to his 0.66% national figure by roughly the ratio of the two exposure measures. That inflation is common to the two regions and cancels from the differential by Corollary 3.

4.9 Wage gradients: propagation at the worker level

The wage regression is the third propagation demonstration where exposure enters Equation (11) as a regressor measured with error whose variance the calibration of Section 3.3 supplies. The question here is not whether the gradient is attenuated but whether the attenuation can be bounded at the worker level, and what remains of the Scotland interaction once it is.

Table 8: Projected ten-year TFP contribution and the Scotland–rUK differential

	κ	rUK	Scotland	Differential
<i>Panel A. Composite exposure index, by cost-saving scenario</i>				
Low	0.110	1.687	1.665	−0.022
Central	0.144	2.209	2.180	−0.029
High	0.182	2.792	2.756	−0.036
<i>Panel B. Single-channel projections, central scenario</i>				
Complementarity only	0.144	2.108	2.077	−0.032
Substitution only	0.144	1.280	1.268	−0.012
<i>Panel C. Exposed task-content share $\text{Sub}_k + \text{Comp}_k$, central scenario</i>				
Thresholds (60, 30)	0.144	3.030	3.009	−0.021
Thresholds (70, 40)	0.144	2.761	2.722	−0.040
Thresholds (80, 50)	0.144	2.548	2.498	−0.050

Notes: Cumulative contribution to total factor productivity over the ten-year horizon, in per cent; the differential is Scotland minus rUK in percentage points. κ is the average cost saving on an exposed task. Panel B rows are alternative projections run on the two channel indices. Panel C replaces $\bar{E}_k$ with the importance-weighted share of task content classified as exposed, at each threshold pair. The band carried into Table 10 is the Panel A range.

Higher-exposure occupations pay markedly more, reflecting how exposure aligns with professional and cognitive task content. The gradient is 1.276 (SE 0.170), so each ten index points of exposure carries about 13.6% higher pay, and pay differs by 95.6% across the observed range of 31.7 to 84.3 index points. The Scotland interaction is −0.300 with a standard error of 0.065 when errors are clustered by occupation, the level at which the exposure regressor varies.

A level control cannot purge either of two confounds sitting behind the interaction. The first is composition. Exposure aligns with graduate, professional and public-sector work, so if Scotland’s within-occupation qualification or sector profile moves with exposure, the interaction absorbs that drift too. Adding highest qualification and a public-sector indicator (Table 9, column 2) cuts the exposure gradient from 1.276 to 0.941 but barely touches the Scotland interaction. Importantly this stability implies Scotland’s flatter gradient is not a proxy for its qualification mix or its larger public sector, since both work through the level of the wage schedule rather than its slope. The gradient’s fall is a different story, though. Qualification is partly a channel through which exposure raises pay, not simply a confound, so column 2 reports a within-qualification return to exposure.

The second confound is London, because β_3 is otherwise identified against an rUK in which the most exposed professional occupations also carry the capital’s pay premium. Absorbing all regional wage levels in fixed effects (Column 3) leaves the slopes and gives −0.216 (SE 0.056), removing about a fifth of the controlled interaction. Giving London its own level and its own exposure slope instead (Column 4), gives −0.187 (SE 0.058), removing about three-tenths. So roughly a third of the controlled interaction is the capital, split between its wage level and its steeper gradient, and the remainder belongs to Scotland.

Against the ex-London comparator, then, the Scottish return to exposure is roughly a fifth lower than in the rest of the UK (0.697 against 0.884). Set against the Scotland level effect of 0.138, the two schedules rotate rather than shift. Scottish pay carries a premium of about 8% at the least-exposed end of the distribution, closes at 73.8 index points, and is about 2% below

Table 9: Exposure–wage gradient: sample, controls, and the London confound

	(1) Baseline (full)	(2) +Controls	(3) +Region FE	(4) +London slope
Exposure E	1.276*** (0.170)	0.941*** (0.144)	0.913*** (0.143)	0.884*** (0.143)
Exposure $\times$ Scotland	−0.300*** (0.065)	−0.269*** (0.062)	−0.216*** (0.056)	−0.187*** (0.058)
Scotland	0.204*** (0.044)	0.173*** (0.042)		0.138*** (0.040)
Exposure $\times$ London				0.354*** (0.059)
London				−0.118*** (0.039)
Demographic controls ^a	Yes	Yes	Yes	Yes
Added controls ^b		Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Region (GOR) FE			Yes	
Clustering	Occ.	Occ.	Occ.	Occ.
Sample	Full	GOR	GOR	GOR
Observations	176,444	176,024	176,024	176,024
Within R^2	0.174	0.235	0.226	0.243

Dependent variable: log gross hourly pay. Worker-level regressions of Equation (11) on the pooled APS 2022–2025, weighted by the APS income weight. Standard errors, clustered by occupation (the level at which the exposure score varies), in parentheses. Column (1) is the full-sample baseline; columns (2)–(4) add the controls of note b and restrict to observations with an observed regional code, so column (2) is the like-for-like comparator for the London treatments in (3)–(4). The Scotland main effect is absorbed by the regional fixed effects in column (3) and is therefore not separately estimable.

^a Age, age², sex, and full-time/part-time status.

^b Highest qualification (education) and a public-sector indicator.

Standard errors are in parentheses: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

the ex-London comparator at the most exposed end. The crossing point sits inside the observed range, so the flattening is not a statement about the tail alone. The regional tilt is therefore not confined to employment quantities. The price of exposed work is also modestly lower in Scotland, and the two margins point the same way for the fiscal translation of Section 4.10.

The gradient sits where the literature would put it: pay rises with measured exposure across every measurement generation, whether the index is ability-based or LLM-rated (Felten et al. 2021; Eloundou et al. 2024; Henseke et al. 2025; Pizzinelli et al. 2023), though because each index carries its own scale, that agreement establishes sign and composition, not magnitude. No such literature exists for the regional margin. I am aware of no comparable within-country estimate of a regional rotation of the exposure–wage schedule, so the Scotland interaction is anchored against internal benchmarks instead, namely the London slope and the public-sector premium of Cribb et al. (2025).

Two limits on the exercise bear stating. Gross hourly pay is recorded on the APS for employees only, so the estimates describe the employee earnings distribution and exclude the self-employed,

who are disproportionately concentrated in exactly the professional occupations that sit at the top of the exposure range. RETURN TO AND CHECK FOR OTHER PROPAGATION PROCEDURES. And the measurement-error correction turns out not to bind. The reliability complement is small ($\lambda = 0.016$), indicating that between-occupation variation in exposure is much larger than the scoring noise, so the classical errors-in-variables correction changes the exposure coefficient by 1.6%, from 1.276 to 1.297, and the Scotland interaction from -0.300 to -0.305 (Section E). Scoring noise is therefore quantitatively unimportant at the worker level, as Table 5 shows it to be in the aggregate differential; where it does bite is the absolute exposure levels, which is precisely where Proposition 1 offers no protection.

4.10 A stylised fiscal translation

This subsection concludes the chain that Section 1 opened, carrying the projection into the income tax net position. What follows is an accounting translation under stated assumptions, and not a forecast.

Mapping the differential of Table 8 to the budget takes four steps. The first converts total factor productivity into labour productivity. Equation (10) delivers a TFP path, whereas earnings track output per hour, and under constant returns with a capital share of one third a permanent one per cent rise in TFP raises output per hour by $1/(1 - \alpha) = 1.5$ per cent once the capital stock has adjusted (Office for Budget Responsibility 2025). The second applies unit long-run pass-through of labour productivity into real earnings, which spread evenly across the horizon is worth about -0.0043 percentage points of earnings growth a year at the centre. The third converts earnings into revenue. Here the elasticity of non-savings, non-dividend (NSND) revenue to earnings is not unity, because the schedule is progressive: a marginal, permanent shift in the level of earnings changes revenue in the ratio of the income-weighted average marginal rate to the average rate on total income, and frozen higher, advanced and top rate thresholds hold that ratio above one (Scottish Government 2026). I estimate it at 1.92 on the APS earnings distribution under the prevailing Scottish schedule, which sits at the upper end of the 1.7–2.0 range international microsimulation evidence finds for progressive income tax systems generally (Section F.3). Allowing the earnings differential to fall in proportion to occupational exposure, rather than evenly, raises it only to 1.974. The final step is per-capita indexation of the block grant adjustment (Scottish Government and HM Treasury 2023). This makes the exercise a differential one since it is Scotland–rUK per-capita revenue growth, not the level of Scottish revenue, that moves the net position. Because the adjustment is indexed to rUK growth and a Scotland-specific shock leaves rUK untouched, only the Scottish elasticity enters, applied to Scottish NSND revenue of £18.6 billion in 2024–25 (HM Revenue and Customs 2026).

Composing the four gives a net-position sensitivity of about $-\text{£}1.5$ million a year at the centre, and $-\text{£}1.2$ to $-\text{£}1.9$ million across the cost-saving band (Table 10). Against the SFC’s forecast net position of £969 million for 2026–27 (Scottish Fiscal Commission 2026), a central sensitivity of this size is around one-sixth of one per cent, well inside the SFC’s income tax forecast-error ranges, which run to the hundreds of millions. The terminal annual effect, in the tenth year, is of order £15 million, or about 1.6 per cent of next year’s forecast position.

The translation abstracts from one channel I do not quantify. Per-capita indexation turns

Table 10: Stylised translation of the exposure differential into the income tax net position

	Low	Central	High
TFP differential (pp, 10-year cumulative)	−0.022	−0.029	−0.036
Labour productivity differential (pp, 10-year cumulative)	−0.033	−0.043	−0.054
Implied earnings growth differential (pp p.a.)	−0.0033	−0.0043	−0.0054
Implied NSND revenue growth differential (pp p.a.)	−0.0063	−0.0082	−0.0104
Net position sensitivity (£m p.a.)	−1.2	−1.5	−1.9
Memo: terminal annual effect, year ten (£m)	−12	−15	−19
Memo: cumulative effect over the ten years (£m)	−64	−84	−107
Memo: forecast net position 2026–27 (£m)		969	

Notes: The TFP differential is the ten-year cumulative Scotland–rUK differential from the composite projection of Section 4.8. The low and high columns are the cost-saving scenarios of that projection, obtained by moving the average cost saving on an exposed task, κ , across its low, central and high settings. The labour productivity row applies $1/(1 - \alpha)$ with a capital share of $\alpha = 1/3$, the conversion from TFP to output per hour under constant returns once the capital stock has adjusted (Office for Budget Responsibility 2025). The earnings row applies unit long-run pass-through of labour productivity into real earnings, annualised over the horizon. The revenue row applies an elasticity of NSND revenue to earnings of 1.92, computed as the ratio of the income-weighted average marginal rate to the average rate on total income across the pooled APS earnings distribution under the prevailing Scottish schedule (Section F.3); frozen higher, advanced and top rate thresholds are what hold it above one (Scottish Government 2026). The net position sensitivity is the revenue growth differential times Scottish NSND revenue of approximately £18.6 billion in 2024–25 (HM Revenue and Customs 2026), under the per-capita indexation of the block grant adjustment (Scottish Government and HM Treasury 2023). The terminal memo is the annual flow in the tenth year, being ten years of accumulated differential growth applied once; the cumulative memo sums the annual flows over the horizon. Neither is an accumulation of the other. The translation abstracts from the population growth differential. The memo net position is the SFC January 2026 forecast (Scottish Fiscal Commission 2026).

on relative population growth, and in the year to mid-2024 Scotland’s population grew by 0.7 per cent against 1.2 per cent in England and 1.1 per cent across the UK as a whole (Park 2023). A demographic differential of that order is two orders of magnitude larger than the earnings differential traced here, and moves the net position through the same arithmetic.

The budgetary significance of the compositional differential comes less from its annual size than from its persistence. Scottish income tax outturn is reconciled against forecast three years after the event, and the reconciliation is applied to the budget in the year it lands (Scottish Government and HM Treasury 2023). A differential driven by occupational structure is persistent rather than transitory, so unlike a sampling error it does not wash out across successive cycles: it enters each year’s forecast, is carried into that year’s reconciliation, and is applied *ex post* against a budget already set three years earlier. The level effect therefore accumulates even though the annual increment does not grow, which is why the ten-year cumulative cost in Table 10 is an order of magnitude larger than any single year’s, and why it arrives with a three-year lag. Accordingly, a forecasting body has to get the direction and the persistence right since the index-point magnitude that Section 4.7 shows to be the least robust part of the measurement matters far less. Getting the sign wrong, and only correcting it three years on, costs more than the annual figure in Table 10 suggests, even though that figure alone would not justify an adjustment today.

References

- Acemoglu, D. (2025), ‘The simple macroeconomics of ai’, *Economic Policy* **40**(121), 13–58.
- Acemoglu, D. and Restrepo, P. (2019), ‘Automation and new tasks: How technology displaces and reinstates labor’, *Journal of economic perspectives* **33**(2), 3–30.
- Acemoglu, D. and Restrepo, P. (2020), ‘Robots and jobs: Evidence from us labor markets’, *Journal of political economy* **128**(6), 2188–2244.
- Autor, D. (2024), Applying ai to rebuild middle class jobs, Technical report, National Bureau of Economic Research.
- Autor, D. H. and Handel, M. J. (2013), ‘Putting tasks to the test: Human capital, job tasks, and wages’, *Journal of labor Economics* **31**(S1), S59–S96.
- Autor, D. H., Levy, F. and Murnane, R. J. (2003), ‘The skill content of recent technological change: An empirical exploration’, *The Quarterly journal of economics* **118**(4), 1279–1333.
- Bound, J. and Krueger, A. B. (1991), ‘The extent of measurement error in longitudinal earnings data: Do two wrongs make a right?’, *Journal of labor economics* **9**(1), 1–24.
- Brynjolfsson, E., Li, D. and Raymond, L. (2025), ‘Generative ai at work’, *The Quarterly Journal of Economics* **140**(2), 889–942.
- Brynjolfsson, E., Mitchell, T. and Rock, D. (2018), What can machines learn and what does it mean for occupations and the economy?, in ‘AEA papers and proceedings’, Vol. 108, American Economic Association 2014 Broadway, Suite 305, Nashville, TN 37203, pp. 43–47.
- Cazzaniga, M., Jaumotte, M. F., Li, L., Melina, M. G., Panton, A. J., Pizzinelli, C., Rockall, E. J. and Tavares, M. M. M. (2024), *Gen-AI: Artificial intelligence and the future of work*, International Monetary Fund.
- Clark, C. E. (1961), ‘The greatest of a finite set of random variables’, *Operations Research* **9**(2), 145–162.
- Creedy, J. and Gemmell, N. (2002), ‘The built-in flexibility of income and consumption taxes’, *Journal of Economic Surveys* **16**(4), 509–532.
- Creedy, J. and Gemmell, N. (2004), ‘The income elasticity of tax revenue: Estimates for income and consumption taxes in the united kingdom’, *Fiscal Studies* **25**(1), 55–77.
- Cribb, J., Dominguez, M. and O’Brien, L. (2025), Scottish public sector employment and pay, Technical report, The IFS.
- Crowley, F., Doran, J. and McCann, P. (2021), ‘The vulnerability of european regional labour markets to job automation: the role of agglomeration externalities’, *Regional Studies* **55**(10–11), 1711–1723.

- Eloundou, T., Manning, S., Mishkin, P. and Rock, D. (2024), ‘Gpts are gpts: Labor market impact potential of llms’, *Science* **384**(6702), 1306–1308.
- Felten, E., Raj, M. and Seamans, R. (2021), ‘Occupational, industry, and geographic exposure to artificial intelligence: A novel dataset and its potential uses’, *Strategic management journal* **42**(12), 2195–2217.
- Frey, C. B. and Osborne, M. A. (2017), ‘The future of employment: How susceptible are jobs to computerisation?’, *Technological forecasting and social change* **114**, 254–280.
- García-Miralles, E., Freier, M., Riscado, S., Leventi, C., Mazzon, A., Abela, G., Boyd, L., Brusbarde, B., Cochard, M., Cornille, D. et al. (2026), ‘Fiscal drag in theory and in practice: A european perspective’, *European Economic Review* p. 105275.
- Government, S. (2024), *Economic Overview of the Public Sector in Scotland*, The Scottish Government.
- Handel, M. J. (2012), ‘Trends in job skill demands in oecd countries’, *OECD Social, Employment, and Migration Working Papers* (143), 0_1.
- Henseke, G., Davies, R., Felstead, A., Gallie, D., Green, F. and Zhou, Y. (2025), ‘How exposed are uk jobs to generative ai? developing and applying a novel task-based index’, *arXiv preprint arXiv:2507.22748*.
- HM Revenue and Customs (2026), Scottish income tax outturn statistics: 2024 to 2025, Official statistics, HM Revenue and Customs. Published 9 July 2026.
- House of Commons Library (2024), ‘Scotland’s economy’, *Research Briefing CDP-2024-0126*.
- Hulten, C. R. (1978), ‘Growth accounting with intermediate inputs’, *The Review of Economic Studies* **45**(3), 511–518.
- Immervoll, H. (2005), ‘Falling up the stairs: The effects of “bracket creep” on household incomes’, *Review of Income and Wealth* **51**(1), 37–62.
- Johnson, P. and Lambert, P. (1989), ‘Measuring the responsiveness of income tax revenue to income growth: a review and some uk values’, *Fiscal Studies* **10**(4), 1–18.
- Moulton, B. R. (1990), ‘An illustration of a pitfall in estimating the effects of aggregate variables on micro units’, *The Review of Economics and Statistics* **72**(2), 334–338.
- Noy, S. and Zhang, W. (2023), ‘Experimental evidence on the productivity effects of generative artificial intelligence’, *Science* **381**(6654), 187–192.
- OECD (2024), *Job Creation and Local Economic Development 2024: The Geography of Generative AI*, OECD Publishing.
- Office for Budget Responsibility (2025), ‘Briefing Paper No. 9: Forecasting productivity’.
- Overman, H. G. and Xu, X. (2022), ‘Spatial disparities across labour markets’, *IFS Deaton Review of Inequalities*.

- Park, N. (2023), ‘Population estimates for the uk, england, wales, scotland and northern ireland: mid’, *Office for National Statistics* pp. 1–9.
- Pizzinelli, C., Panton, A. J., Tavares, M. M. M., Cazzaniga, M. and Li, L. (2023), *Labor market exposure to AI: Cross-country differences and distributional implications*, International Monetary Fund.
- Price, R. W., Dang, T.-T. and Botev, J. (2015), ‘Adjusting fiscal balances for the business cycle: New tax and expenditure elasticity estimates for oecd countries’, *OECD Economic Department Working Papers* (1275), 0_1.
- Scottish Fiscal Commission (2026), Scotland’s economic and fiscal forecasts – january 2026, Technical report, Scottish Fiscal Commission, Edinburgh.
URL: <https://fiscalcommission.scot/publications/scotlands-economic-and-fiscal-forecasts-january-2026/>
- Scottish Government (2026), Scottish budget 2026 to 2027: Scottish tax ready reckoners, Research and analysis, Scottish Government, Edinburgh. ISBN 978-1-80643-637-8. Published 13 January 2026.
- Scottish Government and HM Treasury (2023), ‘Fiscal framework: agreement between the scottish and uk governments’.
URL: <https://www.gov.scot/publications/fiscal-framework-agreement-between-scottish-uk-governments/>
- Scottish Parliament (2016), ‘Scottish fiscal commission act 2016 (asp 17)’. Section 2: duty to prepare forecasts and assessments to inform the Scottish budget.
URL: <https://www.legislation.gov.uk/asp/2016/17>
- Sharps, S., Smith, T., Browne, J., Large, O., Subramanya, R., Tay, P., Ellina, D., Atkinson, I., Lythgow, J. and Muralidharan, R. (2024), ‘The impact of ai on the labour market’, *Institute. global* pp. 1–88.
- Webb, M. (2019), ‘The impact of artificial intelligence on the labor market’, *Available at SSRN 3482150*.
- Williams, D., Tsoukalas, J., Wessels, B. and Gaska, P. (2025), *Scotland’s Productivity Challenge: Exploring the issues-2025*, The Productivity Institute.
- Wooldridge, J. M. (2010), *Econometric Analysis of Cross Section and Panel Data*, 2 edn, MIT Press, Cambridge, MA.
- Yin, M., Vu, H. and Persico, C. (2026), ‘How (un) stable are llm occupational exposure scores? evidence from multi-model replication’, *NBER Working Paper* (w35110).

A Proofs and Derivations

A.1 Common-mode cancellation (Proposition 1)

Proof. Recall the model- M continuous region index

$$\bar{E}_r^M = \frac{1}{100} \sum_k \eta_{k,r} E_k^M, \quad E_k^M = E_k^\star + b_k^M, \quad (12)$$

where $E_k^\star$ is the consensus exposure of minor group k and b_k^M the model- M bias. Substituting the second expression into the first and separating the consensus and bias components,

$$\bar{E}_r^M = \frac{1}{100} \sum_k \eta_{k,r} E_k^\star + \frac{1}{100} \sum_k \eta_{k,r} b_k^M = \bar{E}_r^\star + \frac{1}{100} \sum_k \eta_{k,r} b_k^M. \quad (13)$$

Differencing Equation (13) across regions $r \in \{\text{Scot}, \text{rUK}\}$, the consensus parts collapse to $\Delta \bar{E}^\star = \bar{E}_{\text{Scot}}^\star - \bar{E}_{\text{rUK}}^\star$ and the bias parts to

$$\Delta \bar{E}^M - \Delta \bar{E}^\star = \frac{1}{100} \sum_k (\eta_{k,\text{Scot}} - \eta_{k,\text{rUK}}) b_k^M = \frac{1}{100} \sum_k \Delta \eta_k b_k^M, \quad (14)$$

which is Equation (6). The key step is that the employment shares are proper weights in each region, $\sum_k \eta_{k,r} = 1$, so that the share gaps sum to zero:

$$\sum_k \Delta \eta_k = \sum_k \eta_{k,\text{Scot}} - \sum_k \eta_{k,\text{rUK}} = 1 - 1 = 0. \quad (15)$$

Because of Equation (15), adding or subtracting any occupation-invariant constant $\bar{b}^M$ inside the sum in Equation (14) leaves it unchanged:

$$\frac{1}{100} \sum_k \Delta \eta_k b_k^M = \frac{1}{100} \sum_k \Delta \eta_k (b_k^M - \bar{b}^M) + \underbrace{\frac{\bar{b}^M}{100} \sum_k \Delta \eta_k}_{=0} = \frac{1}{100} \sum_k \Delta \eta_k (b_k^M - \bar{b}^M). \quad (16)$$

Taking $\bar{b}^M$ to be the mean model bias, Equation (16) shows the differential error to be the covariance between the cross-region share gap and the demeaned bias. Part (i) of Proposition 1 follows on setting $b_k^M = \bar{b}^M$ for all k , whereupon $b_k^M - \bar{b}^M = 0$ and the right-hand side vanishes; the absolute levels of Equation (13) remain contaminated by $\frac{1}{100} \sum_k \eta_{k,r} b_k^M$, of the order of the mean bias. Part (ii) is Equation (16) itself. $\square$

A.2 Cross-model stability (Corollary 1)

Applying Equation (14) to two models M and M' and differencing the two error expressions, the consensus benchmark $\Delta \bar{E}^\star$ cancels, leaving

$$\Delta \bar{E}^M - \Delta \bar{E}^{M'} = \frac{1}{100} \sum_k \Delta \eta_k (b_k^M - b_k^{M'}). \quad (17)$$

By the same zero-sum argument of Equation (15), Equation (17) is invariant to a common level shift in either model's bias and is therefore zero whenever the cross-model bias difference $b_k^M - b_k^{M'}$ is constant in k . More generally it is bounded, by the Cauchy–Schwarz inequality, by

$$|\Delta \bar{E}^M - \Delta \bar{E}^{M'}| \leq \frac{1}{100} \|\Delta \eta\|_2 \|b^M - b^{M'}\|_2, \quad (18)$$

so that the cross-model movement in the differential is small whenever the share gap and the bias difference are weakly aligned, even when the bias difference itself is large. This is the formal content of the claim that the differential is more stable across models than the levels, and together Equations (17) and (18) prove Corollary 1.

A.3 Affine bias and scale-free differentials (Corollary 2)

Suppose model M is allowed three ways of being wrong at once: it may shift the origin of the scale, stretch or compress it, and on top of that mis-score individual minor groups. Formally, let $E_k^M = a^M + c^M E_k^* + b_k^M$ with $c^M > 0$. Substituting into the region index and differencing as in Equations (13) and (14),

$$\Delta \bar{E}^M = \frac{1}{100} \sum_k \Delta \eta_k (a^M + c^M E_k^* + b_k^M) = \underbrace{\frac{a^M}{100} \sum_k \Delta \eta_k + c^M \Delta \bar{E}^*}_{=0} + \frac{1}{100} \sum_k \Delta \eta_k b_k^M, \quad (19)$$

which is Equation (8). The intercept vanishes by Equation (15), exactly as the mean bias does in Proposition 1: an affine model's additive part is a common-mode error and the differential is blind to it. The slope is not. It multiplies the differential, so a model that uses less of the $[0, 100]$ scale returns a proportionally smaller gap in index points even where it agrees perfectly about which occupations are the exposed ones. This is a failure of Equation (5) rather than of Proposition 1: a multiplicative disagreement is simply not representable as $\theta_k^* + b_k^M$ with b_k^M constant, so it appears in Equation (6) as a bias that co-varies with the exposure level and therefore, in part, with the share gap.

Part (ii) follows because the transformation is applied to the score distribution, which is common to the two regions. Writing SD^M for the employment-weighted standard deviation of $\{E_k^M\}$ across the UK, a compression by c^M shrinks the gap and the dispersion in the same proportion. So when $b_k^M \equiv 0$ we have $\text{SD}^M = c^M \text{SD}^*$, and the two factors cancel:

$$\frac{\Delta \bar{E}^M}{\text{SD}^M} = \frac{c^M \Delta \bar{E}^*}{c^M \text{SD}^*} = \frac{\Delta \bar{E}^*}{\text{SD}^*}, \quad (20)$$

and neither a^M nor c^M survives. With $b_k^M \neq 0$ the standardised differential retains $(\frac{1}{100} \sum_k \Delta \eta_k b_k^M) / \text{SD}^M$, which is the non-affine residual and is what Table 7 measures.

Part (iii) is immediate. A percentile rank depends on the scores only through their ordering, and a strictly increasing map, of which $E \mapsto a^M + c^M E$ with $c^M > 0$ is one, cannot change an ordering. The percentile differential is therefore untouched by any monotone rescaling, not merely an affine one. The price is that it throws away the cardinal information the index-point gap carries: it can say Scotland is two percentile points down, but not that the gap is twice as

large as some other gap.

Both transformations act on the score distribution and leave the employment shares alone, so neither introduces any regional variation of its own, $\sum_k \Delta\eta_k = 0$ still holds, and the demeaning argument of Equation (16) carries over to the transformed scores unchanged.

A.4 Downstream linear functionals

Corollary 3 (Downstream linear functionals). *Let $F_r^M = \sum_k \eta_{k,r} c_k E_k^M$ with region-common coefficients $c_k \geq 0$. Then*

$$\Delta F^M - \Delta F^\star = \sum_k \Delta\eta_k c_k b_k^M, \quad (21)$$

and under constant coefficients $c_k = c$ the cancellation of Proposition 1 is exact.

Proof. Substituting $E_k^M = E_k^\star + b_k^M$ and differencing across regions gives Equation (21). Unlike the unweighted case of Equation (15), the weighted share gaps need not sum to zero: $\sum_k \Delta\eta_k c_k = \bar{C}_{\text{Scot}} - \bar{C}_{\text{rUK}}$ with $\bar{C}_r = \sum_k \eta_{k,r} c_k$. Under a pure level shift $b_k^M = b^M$, Equation (21) therefore reduces to $b^M(\bar{C}_{\text{Scot}} - \bar{C}_{\text{rUK}})$: the contamination is the level shift scaled by the cross-region difference in average coefficient loading. When the coefficients are constant, $c_k = c$, the sum collapses to $c b^M \sum_k \Delta\eta_k = 0$ by Equation (15), and the cancellation is exact. $\square$

The continuous exposure gap is the special case $c_k = 1/100$; the baseline TFP differential of Section 3.4 is the special case $c_k = \kappa\pi_k$ with $\pi_k = \pi$ constant across occupations, which is what makes the cancellation there exact.

A.5 Errors-in-variables attenuation (Section 3.4)

Consider the conditional wage regression Equation (11) in the single-regressor form $\ln w_i = \alpha + \beta_1 E_{j(i)} + \nu_i$, where the true continuous exposure $E_{j(i)} \in [0, 1]$ is observed with classical within-model error $\tilde{E}_{j(i)} = E_{j(i)} + u_i$, $u_i \sim (0, \sigma_u^2)$ independent of $E_{j(i)}$ and ν_i . The within-model scoring variance is estimated on the $[0, 100]$ scale as $\hat{\sigma}_{g(j)}^2$, which rescaling to the $[0, 1]$ index gives $\sigma_u^2 = \hat{\sigma}_{g(j)}^2/10^4$. The OLS estimator of β_1 using $\tilde{E}$ has probability limit

$$\text{plim } \hat{\beta}_1^{\text{OLS}} = \beta_1 \cdot \frac{\text{Var}(E)}{\text{Var}(E) + \sigma_u^2} = \beta_1(1 - \lambda), \quad \lambda \equiv \frac{\sigma_u^2}{\text{Var}(E) + \sigma_u^2}, \quad (22)$$

the standard attenuation result for classical measurement error in a single regressor (Wooldridge 2010, ch. 4), where λ is the reliability complement and the survey-data context is that of Bound and Krueger (1991).

$$\hat{\beta}_1^{\text{EIV}} = \frac{\hat{\beta}_1^{\text{OLS}}}{1 - \hat{\lambda}}, \quad \hat{\lambda} = \frac{\hat{\sigma}_u^2}{\widehat{\text{Var}}(\tilde{E})}, \quad (23)$$

with standard errors obtained by the delta method on the smooth function $(\hat{\beta}_1^{\text{OLS}}, \hat{\lambda}) \mapsto \hat{\beta}_1^{\text{OLS}}/(1 - \hat{\lambda})$. The correction to the Scotland interaction β_3 in the full specification Equation (11) is treated as approximate, since the interaction regressor $E_{j(i)} \times \text{Scot}_i$ inherits the measurement error of $E_{j(i)}$ only on the Scottish subsample and the resulting attenuation is no longer a single scalar factor.

A.6 Correlated within-family errors in the Monte Carlo

The procedure of Section 3.3 draws $s_j^{(r)} \sim N(s_j, \hat{\sigma}_{g(j)}^2)$ across occupations j , independently under $\rho = 0$ and with an equicorrelated structure within each occupational family otherwise. Consider the contribution of a single family g containing n_g occupations to an employment-weighted aggregate, and let $W_g = \sum_{k \in g} \eta_{k,r}$ be the employment weight the family carries. Under independent draws the family’s occupation-level errors average, so its contribution to the variance of the aggregate is $W_g^2 \hat{\sigma}_g^2 / n_g$. Imposing $\text{Cov}(\varepsilon_i, \varepsilon_{i'}) = \rho \hat{\sigma}_g^2$ for $i \neq i'$ within the family gives

$$\text{Var}(\bar{E}_r) = \sum_g \frac{W_g^2 \hat{\sigma}_g^2}{n_g} \underbrace{[1 + (n_g - 1)\rho]}_{\text{design effect}}, \quad (24)$$

which reduces to the independent case at $\rho = 0$ and rises to $W_g^2 \hat{\sigma}_g^2$ at $\rho = 1$, where the family moves as one. Equation (24) is an exact expression under equicorrelation rather than a bound: the bracket is the familiar design-effect factor, and it exceeds one for any $\rho > 0$, so positively correlated framing errors widen the interval relative to the independent baseline. Because ρ is not identified from four scoring conditions, I do not estimate it; I report the propagation across $\rho \in \{0, 0.2, 0.4\}$ and read the spread across those rows as the sensitivity of the reported uncertainty to the assumption. The same factor applies to the differential, but there it multiplies a variance that Proposition 1 has already made small, which is why the ρ rows of Table 5 move the scoring component visibly and the composed interval hardly at all.

B Data Construction and Coverage

B.1 Crosswalk from O*NET–SOC to UK SOC 2020

The exposure scores are indexed by O*NET–SOC occupation and the employment weights by UK SOC 2020, so the two are bridged by a published concordance that maps O*NET–SOC 2019 codes to UK SOC 2020 unit groups. I work at the three-digit minor-group level. This trades a degree of occupational resolution for cell sizes large enough to support reliable regional employment shares once survey waves are pooled, which is the binding constraint for Scotland, where finer cells thin out quickly. The coarser working level also makes the region-invariance assumption of Assumption 1 more comfortable, since averaging over a minor group smooths the fine distinctions along which any residual regional difference in task content would appear.

Aggregation rule. The concordance is many-to-many, so an aggregation rule is required. Each UK SOC 2020 unit group receives the mean of the scores of the O*NET occupations mapping into it, weighted by the concordance weights where the published file supplies them and unweighted otherwise. Unit groups then aggregate to the three-digit minor group by unweighted mean, since the public APS resolves employment only to the minor group and no domestic weight exists below it. A consequence of the rule is that the unweighted second step gives a thinly populated unit group the same voice as a dominant one. I therefore recompute the differential under two alternative rules, equal weights at both steps, and mapping-mass weights at the second step, which Table 11 reports alongside the baseline. Ultimately, weighting the second step by UK

unit-group employment would be the strongest approach, but it requires the secure-access APS and so is saved for future work granted such access.

Table 11: Sensitivity of the differential to the crosswalk aggregation rule

Aggregation rule	Scotland	rUK	Gap
Baseline (weighted, then unweighted)	0.6582	0.6669	-0.008633
Equal weights at both steps	0.6582	0.6669	-0.008633
Mapping-mass weights at step 2	0.6595	0.6685	-0.008988

Notes: Continuous composite index under the maximum operator, in index points. Step 1 maps O*NET–SOC occupations to UK SOC 2020 unit groups; step 2 maps unit groups to three-digit minor groups. The baseline row reproduces the headline gap of Table 2 by construction.

B.2 The differential at the four-digit level

The three-digit working level compresses the compositional gap for the same reason coarse geographies compress spatial variation, as Section 4.2 notes: any Scotland–rUK sorting across unit groups within a minor group is averaged away before the differential is formed. The public APS resolves occupation only to the minor group, so the comparison below requires the secure-access ASHE data for extension.

B.3 APS pooling and cell-count diagnostics

I pool four APS waves, 2022–2025, applying the survey’s person weights. Pooling is necessary because a single wave leaves too many Scottish minor-group cells sparsely populated for the employment shares to be precise, and the window is short enough that occupational structure is stable across it. The residual thinness is worth recording, because it is the source of the sampling error that dominates the reported interval. 23 of the 104 pooled Scottish occupation cells contain fewer than 100 respondents across the four waves, and the smallest contains 11.

Two design features the bootstrap does not capture. Consecutive APS annual datasets are not independent samples: the APS is assembled from overlapping Labour Force Survey waves, so a respondent can appear in adjacent annual datasets, and a person-level bootstrap of the pooled file treats those appearances as independent draws while also ignoring the survey’s clustered design. Both simplifications understate the sampling variance of the shares, so the composed interval of Table 5 understates the sampling component and should not be read as an exact coverage statement. Separately, the pooled window coincides with the 2022–24 fall in LFS response rates and the associated reweighting, whose limitations for sub-national estimates the ONS itself has flagged; the bootstrap captures sampling variance but not differential non-response between Scotland and the rUK, which would bias the shares rather than widen their interval. The employer-reported ASHE dataset, available under a Data Access Agreement, is the natural external check on the pooled occupational shares, and I again leave that comparison to the secure-access data extension.

Assumption 1 fails on the industry margin exactly where the within-occupation industry mix differs across the border in a way correlated with exposure, and that much is checkable on the

APS, which records industry alongside occupation. For each minor group k I compute the total-variation distance between the Scottish and rUK industry distributions within the occupation, i.e. the share of within-occupation employment that would have to move between industries to equalise the two distributions,

$$\text{TV}_k = \frac{1}{2} \sum_i |\Pr(i \mid k, \text{Scot}) - \Pr(i \mid k, \text{rUK})|,$$

and summarise it in ?? ... ?

Table 12: Industry-mix and within-industry contributions by SIC section

Industry	Share gap (Scot - rUK, pp)	Within-industry exposure gap	Industry-mix contribution	Within-industry contribution
Extraterritorial bodies	-0.0004503	-4.656	-0.00002685	-0.009024
Household employers	-0.0504	-2.141	0.008661	-0.002386
Water & waste	0.1663	-1.939	-0.002096	-0.01772
Arts & recreation	0.1401	-1.372	-0.001102	-0.03904
Health & social work	1.364	-1.315	-0.02811	-0.1998
Professional & technical	-1.706	-1.281	-0.1485	-0.09176
Accommodation & food	0.9293	-1.16	-0.1014	-0.06816
Manufacturing	-1.726	-0.9774	0.002333	-0.06307
Other services	0.08662	-0.9615	-0.005363	-0.02747
Wholesale & retail	-0.2212	-0.9404	-0.00434	-0.09922
Transport & storage	-0.6101	-0.9088	0.01466	-0.0384
Administrative & support	-0.35	-0.8257	0.01769	-0.03341
Public admin & defence	1.532	-0.5018	0.08306	-0.04856
Construction	-0.4983	-0.3656	0.04711	-0.02206
Information & communication	-1.414	-0.3334	-0.1446	-0.01231
Real estate	-0.1885	-0.1449	-0.01189	-0.001545
Finance & insurance	0.173	-0.08175	0.02152	-0.003637
Education	-0.5587	0.7672	0.02436	0.07808
Agriculture, forestry & fishing	0.5335	0.8904	-0.06986	0.0122
Mining & quarrying	1.732	0.9175	0.05923	0.01771
Energy supply	0.6671	1.145	0.02925	0.01348

Notes: Per-section terms summing to the totals of Table 4, ordering A (rUK within-industry exposure, Scottish share differences); ordering B is materially similar section-by-section and its aggregate is reported in Table 4. The industry-mix term is demeaned, $\Delta\sigma_i(\bar{E}_{i,\text{rUK}} - \bar{E}_{\text{rUK}})$, so a section’s entry reflects how unusual its Scotland–rUK share gap is rather than the level of $\bar{E}_{i,\text{rUK}}$; undemeaned terms and both orderings in full are in the online replication material. The industry names are abbreviated from the official SIC 2007 section titles.

C Scoring Protocol and Audit Trail

The regional finding rests on the claim that the only variation across occupations is genuine, not an artefact of a drifting prompt or model. This appendix documents the frozen configuration in enough detail for the exercise to be reproduced or audited independently.

C.1 Frozen configuration

I fix and freeze a single model and prompt before the main run. Decoding is deterministic to the extent the providers allow: the temperature is set to zero and no sampling parameter is

varied across the run, so the only intended source of variation across tasks is the task text itself. Table 13 records it. To make any later change to the elicitation detectable, the scoring script records a SHA-256 hash covering the system message and the user-message template jointly, so the figures either reproduce from the recorded configuration or else fail to match the hash.

Table 13: Frozen scoring configuration

Item	Value
<i>Baseline run</i>	
Provider / model snapshot	Anthropic, <code>claude-sonnet-4-6</code>
Transport	Anthropic Batch API
Run date	8 July 2026
Temperature	0
Maximum output tokens	150
Top- p / top- k	not set (provider default)
Seed	none; the Messages API exposes no seed parameter
Retries / concurrency	2 retries; 8 concurrent requests on the synchronous path
Response format	single JSON object, keys <code>substitution</code> , <code>complementarity</code> , <code>primary_mode</code> , <code>key_factors</code>
Prompt SHA-256	8e831d4ea7c85bb9962c71a7fbf15672bb681b4fb33a61a1df4735c8c6ea175a
V0 batch identifier	msgbatch_01Gvienm9x6cVRoVcpEcP7Qf
<i>Cross-model arm (Section 4.7)</i>	
Alternative provider	OpenAI, <code>gpt-4o-2024-11-20</code> , Batch API; <code>response_format = json_object</code> , same temperature and token cap
Alternative tier	Anthropic, <code>claude-haiku-4-5-20251001</code> , same transport as baseline
<i>Elicitation</i>	
Horizon	10 years, from a base year of 2026
Scoring scale	two integer scores on $[0, 100]$: substitution and complementarity
Criteria	six fixed considerations, presented in a fixed order

C.2 Prompt and rubric

The system message casts the model as an economist assessing what artificial intelligence can do to a stated O*NET task within the horizon, instructs it to assume capability progress continues at approximately the pace observed between 2020 and 2025, and asks for two scores rather than one: a substitution score for the extent to which AI could perform the task end-to-end with no meaningful human involvement, and a complementarity score for the extent to which it could raise the productivity of a worker who is retained. The six considerations the model is directed to weigh cover information-processing content, physical dexterity in unstructured settings, empathy and non-delegable accountability, genuine novelty, verifiability of output, and routinisation. The user message supplies the occupation title and the task text verbatim from O*NET. The full prompt text, the paraphrased variant and the remaining calibration perturbations are reproduced in the replication archive and can be provided on request.

C.3 Completeness of the run

The baseline run scores all 17,945 task–occupation pairs across 894 O*NET occupations, this being the set of task statements carrying an unsuppressed Importance rating in release 30.3. One occupation, 19-4012.01, loses a single task (4.4% of its importance mass) to an unparseable response, and every other occupation is scored in full. The replication archive contains the task-level scores for the baseline and for both alternative models, the calibration variants, the prompt text and its hash, the batch identifiers, the analysis code, and an environment lock file recording the R and package versions used.

D Calibration and Robustness Detail

D.1 The stratified calibration sample

The calibration draws $N = 90$ occupations, eighteen from each of five broad occupational families. The families are obtained by collapsing the O*NET two-digit major groups onto their leading digit, which yields a professional and technical family, a second covering community, legal, education, arts and healthcare practitioner work, a third covering healthcare support, protective, food, building and personal services, a fourth covering sales, office administration, farming, construction and installation, and a fifth covering production and transport. The partition is coarse deliberately. Because $\hat{\sigma}_g^2$ is estimated from four scoring conditions on the occupations drawn within each family, a finer split leaves cells too thin for the variance to be usable, and this is the finest grouping at which every cell carries eighteen sampled occupations. The same $g(j)$ indexes the noise variance throughout Section 3.3 and the propagation of Table 5.

Within each family, ten occupations are drawn at random and eight are the ones sitting nearest the classification boundary. Estimating the framing variance on a purely boundary-proximate sample would bias it upward if ambiguity rises towards the middle of the scale, which is where the boundaries sit. The random stratum avoids this and supplies $\hat{\sigma}_g^2$; the boundary stratum is held out, and is used to check that instability concentrates where the construction says it should rather than to estimate anything.

D.2 Noise versus estimand: why two conditions are excluded from $\hat{\sigma}_g^2$

Of the six scoring conditions, four are meaning-preserving perturbations of the frozen prompt and identify the framing noise. The remaining two, the five-year horizon and the single-composite rubric, ask the model a different question, so I report their occupation-level shifts as named sensitivities and do not pool them into $\hat{\sigma}_g^2$. Pooling them would conflate framing noise with a change of estimand and overstate the variance to which the propagation applies. By Proposition 1, the component of either shift that is common across occupations is in any case innocuous for the differential, which is consistent with the near-unit cross-occupation correlations (0.998 and 0.956) the two variants leave behind.

D.3 Correlated-error propagation

Across $R = 1,000$ draws I perturb each occupation score, $s_j^{(r)} \sim N(s_j, \hat{\sigma}_{g(j)}^2)$ truncated to $[0, 100]$, re-aggregate, re-weight, and record $\Delta \bar{E}^{(r)}$. Because correlated framing errors would otherwise lead independent draws to understate aggregate uncertainty, I allow intra-family correlation $\varepsilon_g^{(r)} \sim N(0, \hat{\sigma}_g^2[(1 - \rho)I + \rho \mathbf{1}\mathbf{1}^\top])$ for $\rho \in \{0, 0.2, 0.4\}$. The resulting inflation is exactly the design-effect factor derived at Equation (24), which reduces to the independent case at $\rho = 0$; since ρ is not identified from four scoring conditions I do not estimate it, and read the spread across the ρ rows of Table 5 as the sensitivity of the reported uncertainty to the assumption.

D.4 The classification stability score

Section 3.3 treats the classified indices separately because they are nonlinear at the thresholds. For each task I ask how many noise standard deviations its scores sit from the nearest classification cut-off, and convert that distance into a probability that the task keeps its class under re-framing. With normalised importance weight $\tilde{\omega}_{j,t}$ and task-level noise standard deviations σ_g^{sub} and σ_g^{comp} estimated from the same four meaning-preserving conditions, the occupation-level stability score is

$$S_j = \sum_{t \in T_j} \tilde{\omega}_{j,t} \Phi \left(\min \left\{ \frac{|s_{j,t}^{\text{sub}} - \theta^{\text{sub}}|}{\sigma_{g(j)}^{\text{sub}}}, \frac{|s_{j,t}^{\text{comp}} - \theta^{\text{comp}}|}{\sigma_{g(j)}^{\text{comp}}} \right\} \right), \quad (25)$$

with Φ the standard normal cdf, so that S_j reads as the importance-weighted probability that an occupation's task content keeps its class. A task exactly on a threshold contributes $\Phi(0) = 0.5$ and one two noise standard deviations clear of it contributes $\Phi(2) \approx 0.98$. The reporting landmark I flag against, $S_j < 0.75$, therefore marks an occupation whose importance-weighted task mass sits on average around two-thirds of a standard deviation from the nearest cut-off, roughly the point at which a task is about as likely as not to be boundary-adjacent.

The landmark never binds. No occupation and no minor group falls below it, so there is no flagged set to compare against an unflagged one and no set of occupations for the headline differential to be reported with and without. What the held-out boundary stratum can be used for is the weaker check reported in Section 4.5: boundary-proximate occupations flip class across the meaning-preserving variants at 55.0% against 6.0% for the randomly drawn ones, so the construction locates instability where it should, and excluding minor groups by average stability leaves the classified gap unchanged to the fourth decimal place at every cutoff up to 0.85. The reason Equation (25) is a null diagnostic is because the occupations which do flip carry small and regionally similar employment shares, is itself part of why the classified gap behaves as Section 4.5 describes. COULD PROBABLY DO AWAY WITH THIS CHECK

D.5 Cross-model arm

The Corollary 1 and Corollary 2 tests are reported in the main text rather than here, because the cross-model result is a finding and not a robustness annexe. Table 6 gives the differential and the mean occupation-level bias under the alternative provider and the alternative tier, together with the correlation of that bias with the cross-region employment-share gap, which is the

empirical analogue of the surviving term in Equation (6). Table 7 gives the same differentials in standard-deviation units and in employment-weighted percentile points, which is where the affine component of the disagreement is removed.

E Additional Results

Measurement-error benchmark for the wage gradient

Exposure enters the wage regression at the SOC minor-group level, so every worker in a minor group carries the same score and the measurement error is grouped and heteroskedastic across occupational families. The classical errors-in-variables formula $\hat{\beta}_1^{\text{EIV}} = \hat{\beta}_1^{\text{OLS}} / (1 - \hat{\lambda})$, with reliability complement $\lambda = \sigma_u^2 / \text{Var}(E_j)$ and $\sigma_u^2 = \hat{\sigma}_{g(j)}^2 / 10^4$, assumes independent homoskedastic error per observation and so mis-states the correction here. The violation runs in the direction that overstates it, so I report the classical figure as an upper benchmark on the attenuation rather than as the correction itself.

The formula is also a single-regressor result, which is why it cannot be read off term by term for the interaction. The product regressor $E_{j(i)} \times \text{Scot}_i$ inherits the measurement error $u_j \text{Scot}_i$, whose variance is σ_u^2 in Scotland and zero in the rest of the UK: no single reliability describes it, and because E and $E \times \text{Scot}$ are correlated the attenuation of β_1 and β_3 is entangled rather than separable. The correction nonetheless follows once β_3 is read as the gap between the rUK exposure slope, β_1 , and the Scottish one, $\beta_1 + \beta_3$. Each is a single mismeasured regressor within its own region and attenuates by that region’s reliability; since σ_u^2 is a property of the shared occupation score and $\text{Var}(E)$ differs across regions only through employment composition, the two reliabilities are equal to a close approximation and the common factor $1/(1 - \hat{\lambda})$ carries the interaction as well. This is the same assumption the β_1 correction already makes, so I apply it to both.

Attenuation under the full specification

Equation (22) computes the reliability on the total cross-occupation variation in E , but Equation (11) conditions on industry and year fixed effects and the worker controls, and those absorb part of that variation. By the Frisch–Waugh–Lovell theorem the coefficient on $\tilde{E}$ equals the simple regression of the residualised outcome on the residualised regressor. The measurement error is orthogonal to the controls by assumption, so residualising $\tilde{E}$ on the control set Z leaves $\varrho + u$, where ϱ is the population residual from projecting E on Z , and applying Equation (22) to that regression gives

$$\text{plim } \hat{\beta}_1^{\text{OLS}} = \beta_1 \cdot \frac{\text{Var}(\varrho)}{\text{Var}(\varrho) + \sigma_u^2}. \quad (26)$$

The effective reliability is therefore computed on the residual rather than the total variation in exposure, and any control that explains part of E strictly worsens the attenuation. This runs against the grouped-error violation noted above, which pulls the other way by overstating the correction, and the two are recorded here because neither is large enough to matter at $\hat{\lambda} = 0.016$: the reported figure is an upper benchmark on an attenuation that is negligible under either departure.

Table 14: Scoring measurement error in the wage gradient: OLS and the classical benchmark

	OLS	EIV benchmark	SE (OLS)
Exposure (β_E)	1.276	1.297	0.170
Exposure $\times$ Scotland ($\beta_{E \times \text{Scot}}$)	-0.300	-0.305	0.065

Notes: Worker-level regression of Equation (11) on the pooled 2022–2025 APS, $N = 176,444$, weighted by the APS income weight, with industry and year fixed effects and standard errors clustered by occupation, the level at which the exposure regressor varies; the occupation $\times$ region figure of 0.226 on the Scotland interaction is reported in Section 4.9 as a comparison. The EIV benchmark applies $1/(1 - \hat{\lambda})$ with $\hat{\lambda} = 0.016$ to both the exposure slope and the interaction, the latter under the equal-reliability argument in the text.

Table 14 reports the benchmark. The reliability complement is small ($\lambda = 0.016$), because the between-occupation variation in exposure dwarfs the scoring noise. The benchmark nudges the exposure slope up by 1.6%, from 1.276 to 1.297, and the interaction by the same factor, from -0.300 to -0.305 . The correction is immaterial even under an estimator whose violated assumptions overstate it, so the grouped structure of the measurement error cannot rescue an attenuation this small.

F Specifications and Projection Detail

F.1 Employment specifications

The first of the three exercises in Section 3.4 expresses the compositional gap of Equation (3) as an estimable gradient. I stack occupation $\times$ region cells and estimate

$$y_{k,r} = \alpha + \beta \bar{E}_k + \delta \text{Scot}_r + \zeta(\bar{E}_k \times \text{Scot}_r) + \phi_{s(k)} + u_{k,r}, \quad (27)$$

weighted by cell employment, where $y_{k,r}$ is log occupational employment and, in a second specification, the employment share; $\phi_{s(k)}$ are one-digit SOC major-group fixed effects; and errors are clustered by occupation. Since $\bar{E}_k$ does not vary by region, the interaction ζ is identified purely from the interaction of the common exposure score with the regional employment distribution, and a negative estimate means composition insulates Scotland.

The estimates agree with the decomposition they restate. With major-group fixed effects, $\hat{\zeta}$ is -0.629 (SE 0.241) for log employment and -0.018 (0.007) for the employment share, implying within broad occupation groups, Scottish employment sits systematically further down the exposure gradient than rUK employment.

F.2 Productivity calibration

Where Equation (10) comes from. The projection rests on an envelope result rather than on an assumed functional form, and deriving it matters here because it is what makes the regional differential a linear functional of the scores and therefore subject to Corollary 3. Let the final good be produced under constant returns from a set of tasks with prices $p_1, \dots, p_N$ and unit cost function $c(p_1, \dots, p_N)$, homogeneous of degree one. By Shephard’s lemma the cost-minimising

task demands are $\partial c/\partial p_n$, so

$$\varsigma_n = \frac{\partial \ln c}{\partial \ln p_n} = \frac{p_n (\partial c/\partial p_n)}{c} \quad (28)$$

is the Shephard cost share of task n , and the shares sum to one by Euler’s theorem under linear homogeneity. If AI lowers the cost of task n by the fraction κ_n , so that $d \ln p_n = -\kappa_n$, then totally differentiating gives $d \ln c = \sum_n \varsigma_n d \ln p_n = -\sum_n \varsigma_n \kappa_n$, and since total factor productivity is the inverse of unit cost at fixed input prices,

$$\Delta \ln \text{TFP} = -d \ln c = \sum_n \varsigma_n \kappa_n. \quad (29)$$

This is Hulten (1978) applied at the task rather than the sector level, which is the step Acemoglu (2025) takes within the task framework of Acemoglu and Restrepo (2019). Aggregating Equation (29) over a region’s employment distribution requires one further step, and it is where the second parameter of Equation (10) enters. Set $\kappa_n = \kappa$ on tasks AI in fact performs and zero otherwise. Exposure identifies the task content AI *could* perform, and only a fraction π of that content is profitably automated over the horizon, so the displaced cost share of minor group k is $\pi \bar{E}_k$ rather than $\bar{E}_k$. Substituting into Equation (29) and weighting by employment gives Equation (10). The two parameters are therefore distinct objects: ς_n in Equation (28) is a Shephard cost share that sums to one across tasks, whereas π in Equation (10) is an automatable fraction bounded in $[0, 1]$ and constant across occupations. Only the latter is calibrated.

The imported parameters. I import deep params, so that the Scottish and rUK paths differ only through their employment-weighted exposure. The exposure measurement is my own and the architecture follows the Office for Budget Responsibility (2025), but the productivity parameters are Acemoglu’s rather than the OBR’s, with one exception noted below. The average cost saving on an exposed task, κ , rests on the two task-level randomised experiments Acemoglu relies on: Noy and Zhang (2023), who find substantial time savings on mid-level professional writing tasks, and Brynjolfsson et al. (2025), who find productivity gains among customer-support agents concentrated among the least experienced. Noy and Zhang (2023) find a 40% labour cost saving on writing tasks and Brynjolfsson et al. (2025) find 14% on customer-support tasks; averaging the two gives labour cost savings of about 27% on the tasks they affect, which Acemoglu (2025) converts to overall cost savings of about 15% using industry labour shares. The low, central and high columns of Table 8 move κ across $\{0.110, 0.144, 0.182\}$, and all three are overall rather than labour cost savings. The central setting is the value implied by Acemoglu’s own arithmetic: $0.0066/(0.199 \times 0.23) = 0.144$, so it reproduces his headline 0.66% at his exposure share and automatable fraction. The low setting is the bottom of his 0.53–0.71% range. The high setting is the Office for Budget Responsibility (2025) figure, which takes a task-level cost saving of 32% and a UK labour share of 0.57, giving $0.32 \times 0.57 = 0.182$; the OBR pairs it with a more generous automatable fraction of 0.33 and an exposure share of 0.40, which I do not adopt, so only the cost-saving parameter is imported here. Applying Acemoglu’s 27% *labour* cost saving directly as κ would double-count the labour share and is not a scenario but a change of units. The automatable fraction is set to $\pi_k = \pi = 0.23$ for every occupation, this being the share of exposed tasks

that Acemoglu judges profitably automatable over the decade. Because π_k does not vary across k , Corollary 3 applies and the regional differential inherits the cancellation of Proposition 1 exactly; the levels of the path, by contrast, are proportional to π and carry its full model dependence. I retain the ten-year horizon. The contribution is the regional differential in the implied path, not a reassessment of Acemoglu’s headline magnitude, which is contested and outside the scope of this dissertation. The reported regional levels (2.21% for rUK, 2.18% for Scotland) sit above his national figure of about 0.66% by construction, because my continuous index is broader than his roughly 20% task-exposure measure, so I place weight on the differential and its sign, and none on the levels.

F.3 The elasticity of NSND revenue to earnings

The third step of Section 4.10 needs the proportional change in Scottish NSND revenue induced by a marginal, permanent proportional change in the level of earnings. No Scotland-specific figure is published, but the object is a standard one. It is what the public finance literature calls the *revenue elasticity* or *tax-to-base elasticity* of a progressive income tax, and it is the formal expression of fiscal drag: a schedule with nominal, unindexed thresholds mechanically raises revenue faster than the tax base whenever earnings grow, because a rising share of income is taxed at higher marginal rates. Creedy and Gemmell (2004) set out the UK approach, building on Creedy and Gemmell (2002) and Johnson and Lambert (1989), and it is a structural property of the schedule at a point in time rather than a parameter to be estimated from a time series. If every taxpayer’s income rises by the fraction δ and the thresholds do not move, taxpayer i ’s liability rises by $m_i y_i \delta$, where m_i is the marginal rate at y_i , so

$$\varepsilon = \frac{\sum_i \omega_i m_i y_i}{\sum_i \omega_i T_i} = \frac{\text{income-weighted average marginal rate}}{\text{average rate on total income}}, \quad (30)$$

with ω_i the APS income weight, T_i the liability under the prevailing Scottish schedule and the personal allowance taper carried explicitly. No assumption about threshold indexation is required, because Equation (30) is evaluated at the schedule as it stands. Progressivity alone puts $\varepsilon > 1$; the frozen higher, advanced and top rate thresholds are what keep it there as earnings grow (Scottish Government 2026).

On the pooled APS the income-weighted average marginal rate is 0.296 against an average rate on total income of 0.154, so $\varepsilon = 1.922$. Progressivity alone therefore very nearly doubles the revenue consequence of a given earnings differential.

García-Miralles et al. (2026) compute the same elasticity by microsimulation for 21 European personal income tax systems and find it clusters between 1.7 and 2.0 in fourteen of them, with an unweighted average close to 1.8; Price et al. (2015) report an OECD average of about 1.85 using representative-household calculations. $\varepsilon = 1.922$ sits inside that range, close to its upper end, which is what a six-band schedule with a tapered personal allowance and three widely-spaced higher bands should be expected to produce: García-Miralles et al. (2026) find that elasticities are driven jointly by bracket progressivity and by the phase-out of allowances and credits, and the Scottish personal allowance taper is exactly such a phase-out, levied at an effective marginal rate one and a half times the headline band rate. Immervoll (2005) found comparatively low bracket-

creep effects for the UK in a study using late-1990s data, but explains this by the contemporary UK schedule's few, wide bands. However, that structure is not the current Scottish one, which has added three further bands since 2018 specifically to increase progressivity at the top of the distribution, so a higher elasticity than Immervoll's UK estimate is the expected consequence of the reform rather than an anomaly. I am not aware of a published UK or Scottish estimate of this elasticity to benchmark against directly. The SFC's own income tax forecast instead grows a taxpayer-level micro-simulation of the Survey of Personal Incomes by projected earnings and re-applies the current schedule taxpayer by taxpayer, which captures fiscal drag endogenously and has no need of a scalar summary of it. Equation (30) is a reduced-form statement of what that exercise does in aggregate, and the absence of a published SFC elasticity reflects that choice of method rather than the absence of the underlying effect.

Two departures from Equation (30) matter, and only one of them moves the number. The first is coverage. The APS records gross pay for employees only, so the computed distribution excludes self-employment and pension income and understates the upper tail, where marginal rates are highest. Grossed to a single wave, the implied aggregate liability is roughly three-fifths of the HMRC outturn, and Table 15 reports the ratio so that ε can be read against it. The direction of the resulting bias is worth being precise about. A fatter upper tail raises the numerator and the denominator of Equation (30) together, and since the marginal rate is bounded above by the top rate while the average rate is not, it raises the denominator proportionally more. The employee-only figure is therefore a mild overstatement, and I read 1.922 as the top of a narrow range rather than a point estimate. Because Equation (30) is a ratio of two identically weighted sums, it is invariant to the scale of the weights, and nothing in it turns on the grossing factor that the coverage row exposes.

The second is incidence. The flat pass-through of Section 4.10 spreads the earnings differential evenly, whereas the mechanism of Equation (10) concentrates it in exposed occupations, which are better paid and face higher marginal rates. Letting worker i 's earnings change in proportion to the exposure of their occupation, and normalising so the aggregate change is unchanged, gives the incidence-weighted elasticity

$$\varepsilon^E = \frac{\sum_i \omega_i m_i y_i E_{j(i)}}{\bar{E}_y \sum_i \omega_i T_i}, \quad \bar{E}_y = \frac{\sum_i \omega_i y_i E_{j(i)}}{\sum_i \omega_i y_i}, \quad (31)$$

which exceeds ε whenever exposure and the marginal rate both rise with earnings, as they do here. The correction is small: $\varepsilon^E = 1.974$ against $\varepsilon = 1.922$, so concentrating the earnings differential on exposed workers rather than spreading it evenly moves the fiscal figure by 2.7%. Section 4.10 therefore carries the flat ε .

Table 15: Elasticity of Scottish NSND revenue to earnings

Quantity	Value
Income-weighted average marginal rate	0.296
Average rate on total income	0.154
Elasticity of NSND revenue to earnings, ϵ^E	1.922
Incidence-weighted elasticity, ϵ^E	1.974
Implied aggregate liability (GBP m)	45786
HMRC outturn, 2024-25 (GBP m)	18635
Coverage ratio (implied / outturn)	2.46
Scottish observations	18,550
Tax year of schedule	2026-27

Notes: Computed by `19_nsnd_elasticity.R` on the pooled APS 2022–2025 Scottish employee earnings distribution, weighted by the APS income weight, under the Scottish schedule of the stated tax year with the personal allowance taper carried explicitly. ϵ is Equation (30) and ϵ^E is Equation (31). The coverage row compares the aggregate liability implied by the weighted APS distribution with the HMRC outturn, and the elasticity should be read against that ratio.